

AIMScribe — Software Requirements Specification

Integration specification for the CMED engineering team
AIMS LAB · Independent University, Bangladesh

Document	AIMS-SRS-001
Version	1.7
Date	9 September 2026
Status	Baseline for integration. Items marked OD-nn are open and need a decision.
Relationship to other documents	Complements <code>CMED_INTEGRATION_README.md</code> (narrative) with numbered, testable requirements.
Agent version	2.3.1
Wire protocol	2
Target deployment	7 clinics · 14 consulting rooms · 30 doctors · 16 enrolled laptops

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Part I — Context

1. Introduction

1.1 Purpose

This document states, in numbered and individually testable form, what the AIMScribe system must do from the moment a microphone opens in a consulting room to the moment a verified recording rests in the AIMS LAB archive.

It exists so that CMED's engineers can build their side of the integration against a fixed, unambiguous contract, and so that both organisations can later point at a specific requirement and say *that one is met* or *that one is not*.

1.2 Scope

In scope. Audio capture on the consulting-room PC; segmentation; local encrypted spooling; the cryptographic evidence chain; upload and server-side verification; session lifecycle including automatic patient-to-patient handover; the consultation gate; the recording-control overlay; device enrolment; recording authorisation; the CMED integration surface; the AIMS LAB backend; the archive worker; and the server capacity needed to run all of it.

Out of scope. The ASR (speech-to-text) and NER (clinical entity extraction) pipeline. Those run entirely inside AIMS LAB, strictly downstream of everything described here, and require nothing whatsoever from CMED. They consume archived audio; they do not participate in producing it. No requirement in this document depends on them, and none is changed by them.

Also out of scope: CMED's own clinical application, its database, its user interface and its authentication. This SRS states what AIMScribe needs to *receive*, never how CMED should build what sends it.

1.3 Intended audience, and how to read it

If you are	Read
A CMED engineer implementing the integration	§4.6, §5.4, §6.1, §6.2, §7.7, Appendix A, Appendix B
A CMED architect or tech lead	§1–§3, §5, §9, §13, §14
A CMED security reviewer	§4, §5, §8.4, §9.4
AIMS LAB engineering	All of it
An infrastructure or procurement decision-maker	§10, §11

The shortest useful summary for CMED: you send two signals to a WebSocket on `127.0.0.1`, and you read the replies. That is the entire integration. Everything else in this document is AIMS LAB's obligation, written down so you can hold us to it.

1.4 Definitions and acronyms

Term	Meaning
Agent	<code>AIMScribe_Agent.exe</code> , the Windows tray application on each consulting-room PC
Backend	The AIMS LAB API service (FastAPI), currently hosted on Render
Archive worker	A process on the AIMS LAB server that copies verified audio to permanent storage
Session	One recorded consultation, identified by a ULID
Segment	One 30–60 s audio clip; the unit of encryption, upload, retry and loss
Spool	The agent's local encrypted store-and-forward buffer
Chain	The per-session Ed25519-signed hash chain proving the recording is complete and unaltered
Grant	A short-lived signed authorisation without which the agent will not record
Enrolment token	A single-use administrator-issued secret that binds one PC to one clinic
Device token	The long-lived bearer credential the backend issues to a PC at enrolment
Trigger	The signal from CMED that a consultation has begun
Flag / arm	The signal from CMED that a prescription has been built, permitting the session to end
Gate	The rule that a new trigger does not end the current recording until the flag has arrived
Overlay	The small always-visible Stop/Pause window shown while recording
DPAPI	Windows Data Protection API, used to wrap secrets at rest on the PC
ULID	Universally Unique Lexicographically Sortable Identifier
PLP	Power-Loss Protection, a property of enterprise SSDs

1.5 References

Ref	Document
R1	<code>CMED_INTEGRATION_README.md</code> — the narrative companion to this SRS
R2	<code>PARTNER_INTEGRATION_REQUIREMENTS.md</code> — the same interface offered to Aalo and Amader Susastho
R3	<code>INTEGRATION_SPECIFICATION.md</code> — the as-built wire contract, protocol 2
R4	<code>ARCHITECTURE.md</code> , <code>SECURITY.md</code> , <code>OPERATIONS.md</code> — AIMS LAB internals
R5	RFC 8032 (Ed25519), RFC 7519 (JWT), RFC 6455 (WebSocket), RFC 5116 (AES-GCM)
R6	W3C <i>Secure Contexts</i> — the rule permitting <code>ws://127.0.0.1</code> from an HTTPS page
R7	ISO/IEC/IEEE 29148:2018 — the requirements conventions this document follows

1.6 Requirement conventions

Every requirement has an identifier of the form `SRS-GRP-nn`, a priority and a verification method. Identifiers are permanent: a withdrawn requirement is marked withdrawn, never reused.

Keywords follow RFC 2119. **Must** / **shall** is binding. **Should** is a strong recommendation whose omission needs a written reason. **May** is genuinely optional.

Priority	Meaning
M	Mandatory. The system is not acceptable without it.
S	Should. Expected at general release; may be deferred through the pilot by agreement.
C	Could. Desirable, scheduled only if it costs little.

Verification	Meaning
T	Automated test
D	Demonstration on a running system
I	Inspection of code, configuration or output
A	Analysis, calculation or measurement

The **Owner** column names who builds it: *AIMS*, *CMED*, or *Joint* where both sides change at once.

2. Overall description

2.1 Product perspective

AIMScribe is an evidence-producing recorder. It is not a clinical system, it holds no clinical record, and it makes no clinical decision.

Its single hard problem is one sentence long: **AIMScribe can hear a consultation but cannot know whose it is.** A microphone captures sound; it does not capture the patient's identity, the doctor's identity, or the fact that a consultation has begun at all. Only the clinical application the doctor is already using knows those things.

That is the whole reason CMED is involved. CMED is not being asked to record anything, store anything, or protect anything. CMED is being asked to say *a consultation with this patient, by this doctor, has just begun* — and later, *it is finished*.

Everything else in this document follows from those two sentences.

2.2 The three identities — the system's central invariant

Almost every design decision below is a consequence of keeping these three apart. Conflating any two of them is what produced misfiled consultations in v1.

Identity	Authoritative source	Changes how often	Can a browser page set it?
Clinic (<code>hospital_id</code>)	The PC's enrolment record, held by the backend	Never, for a given PC	No. Not by any route.
Doctor (<code>doctor_id</code>)	CMED, per consultation	Twice a day — morning and afternoon shifts share a laptop	Only as a <i>claim</i> , which the backend then checks against its own register
Patient (<code>patient_id</code>)	CMED, per consultation	Every consultation	Only as a claim, and never stored as a name

Why the clinic comes from the machine and not from the page. `hospital_id` is the top-level folder name on the archive volume. If a page could set it, a mistyped or malicious value would create a new archive tree and the recording would disappear into it silently. It is fixed at enrolment by an administrator and is not an input to anything afterwards.

Why the doctor cannot come from the machine. A consulting room runs two shifts. Binding the doctor to the PC filed every afternoon consultation under the morning doctor — silently, and in the filename, which is the worst possible place for a wrong value because it looks authoritative. The doctor must therefore come from CMED, which knows the rota because doctors log in there.

SRS-INV-01 [M, I, AIMS] `hospital_id` shall be derived exclusively from the device's enrolment record and shall never be accepted from any client payload, on any interface.

SRS-INV-02 [M, I, AIMS] `hospital_id` shall be immutable for the life of a device enrolment. A clinic's *display name* may be changed at any time; its identifier may not, because existing archive paths depend on it.

SRS-INV-03 [M, T, AIMS] `doctor_id` shall have no default and no fallback anywhere in the chain. A trigger naming no doctor shall be refused rather than attributed to a guess.

SRS-INV-04 [M, T, AIMS] The agent shall accept `patient_name` from the browser for on-screen display only. It shall never be written to the chain, the filename, or the archive path.

2.3 User classes

Class	Count	Technical skill	What they touch
Doctor	~30	Low. Uses CMED all day; has no interest in AIMScribe.	The overlay's two buttons, and nothing else. Never a terminal.
Clinic administrator	~7	Low–moderate	Runs the installer once per PC. Pastes one token.
AIMS LAB operator	2–3	High	Mints enrolment tokens, watches alerts, manages the archive
CMED developer	2–4	High	Implements §6.1; never operates the system
Auditor / researcher	Occasional	Moderate	Verifies a chain after the fact; reads, never writes

SRS-USR-01 [M, I, AIMS] No routine task performed by a doctor or a clinic administrator shall require a command line, a script, a `.bat` file or a terminal window. Clinical machines receive a signed installer and a graphical interface only.

This is not a preference. A `.bat` file on a clinical PC is a support incident waiting to happen and an audit finding waiting to be written.

2.4 Operating environment

Element	Specification
Consulting-room PC	Windows 10 21H2 or Windows 11; x64; ≥ 4 GB RAM; ≥ 60 GB free on the spool volume
Audio input	USB speakerphone or microphone. The capture path is a requirement, not a detail — see §7.1a.
Browser	Chrome or Edge, current stable, on the same PC as the agent
Clinic network	Intermittent by assumption. Broadband where available, mobile tethering elsewhere.
Agent runtime	Python 3.12 packaged with PyInstaller; nothing installed on the PC beyond the app
Backend	FastAPI on Python 3.12; containerised; Render today, portable by requirement
Database	PostgreSQL 15+ (Neon today)
Object storage	S3-compatible (Cloudflare R2 or MinIO)
Archive	AIMS LAB server, <code>D:\AIMSLAB_AUDIO_STORAGE</code>
Locale	<code>Asia/Dhaka</code> (UTC+06). All wire timestamps are RFC 3339 with an explicit offset.

2.5 Design and implementation constraints

ID	Constraint	Consequence
CON-01	CMED will not change its architecture for AIMScribe	The integration must be additive: a few lines of JavaScript on pages that already exist
CON-02	CMED exposes a read API and accepts no writes from us	AIMS LAB never calls CMED. All traffic is CMED → agent, on the same machine.
CON-03	CMED will not hold, generate or rotate a cryptographic key	Grant minting moves to the AIMS LAB backend (§5)
CON-04	Clinical PCs get an installer, never scripts	<code>SRS-USR-01</code>
CON-05	Both source repositories are public	No secret of any kind may appear in either. Token sheets are credentials.
CON-06	Clinic bandwidth is limited and unreliable	Store-and-forward is mandatory, not an optimisation
CON-07	The archive is evidence	Lossy re-encoding is prohibited; a re-encoded file cannot be hashed back to what was recorded
CON-08	Consultations run back to back, 20–30 s apart	Session handover must be seamless; no start-up cost may fall between two patients
CON-09	The browser page is untrusted input	Not because CMED is distrusted as an organisation, but because a page is a page and anything on the PC can open one

2.6 Assumptions and dependencies

ID	Assumption	If it proves false
ASM-01	The doctor's browser runs on the same physical PC as the agent	The loopback integration cannot work; a fundamentally different design is required
ASM-02	CMED can execute JavaScript on the patient-details page	No integration is possible without it
ASM-03	CMED holds a stable <code>patient_id</code> , <code>doctor_id</code> and a per-clinic identifier	These are the trigger; without them nothing can be filed
ASM-04	A "build prescription" action exists and is performed for every patient	The gate (§7.7) needs a signal; §11 covers its absence
ASM-05	Patient consent is obtained by clinic process before the consultation	Consent is enforced technically in three places but originates in process
ASM-06	CMED authenticates its own doctors	AIMScribe does not authenticate anyone against CMED and never will
ASM-07	PC clocks are within ± 5 min of true time	Grant validation fails; see §11

2.7 Division of responsibility

Concern	AIMS LAB	CMED
Audio capture, segmentation, encryption	•	
Cryptographic keys — all of them	•	
Device enrolment and its tokens	•	
Grant minting and verification	•	
Backend, database, storage, archive	•	
The recording-control overlay	•	
Agent installation and support	•	
Server capacity and cost	•	
Sending the consultation trigger		•
Sending the prescription-built flag		•
Reading the reply and failing quietly		•
Giving us the exact origin for the allowlist		•
Identifier stability over time		•
The trigger payload's shape	●	●

CMED's total build is four rows. §13.2 estimates it at 2–4 developer-days.

3. System architecture

3.1 Component inventory

#	Component	Runs on	Language / stack	Talks to
1	CMED web page	Doctor's browser	CMED's stack + ~60 lines of JS	The agent, over loopback WebSocket
2	Agent — control API	Consulting-room PC	FastAPI on 127.0.0.1:5050	The page (in), the controller (out)
3	Agent — capture	Same process, own thread	PyAudio	The sound card
4	Agent — segmenter	Same process, own thread	NumPy RMS + ZCR	Capture queue → spool
5	Agent — spool	Same process	AES-256-GCM + journal	Local disk
6	Agent — chain	Same process	Ed25519	Spool, backend
7	Agent — uploader	Same process, async	aiohttp	Backend, object storage
8	Agent — overlay	Same process, UI thread	tkinter	The doctor
9	Backend API	Render or AIMS LAB	FastAPI	Agents, worker, admins
10	Database	Neon / self-hosted	PostgreSQL 15+	Backend only
11	Object storage	R2 / MinIO	S3 API	Backend, agent (presigned), worker
12	Archive worker	AIMS LAB server	Python, outbound-only	Backend, archive volume
13	<i>ASR / NER pipeline</i>	AIMS LAB	<i>out of scope</i>	Reads the archive

3.2 Deployment topology

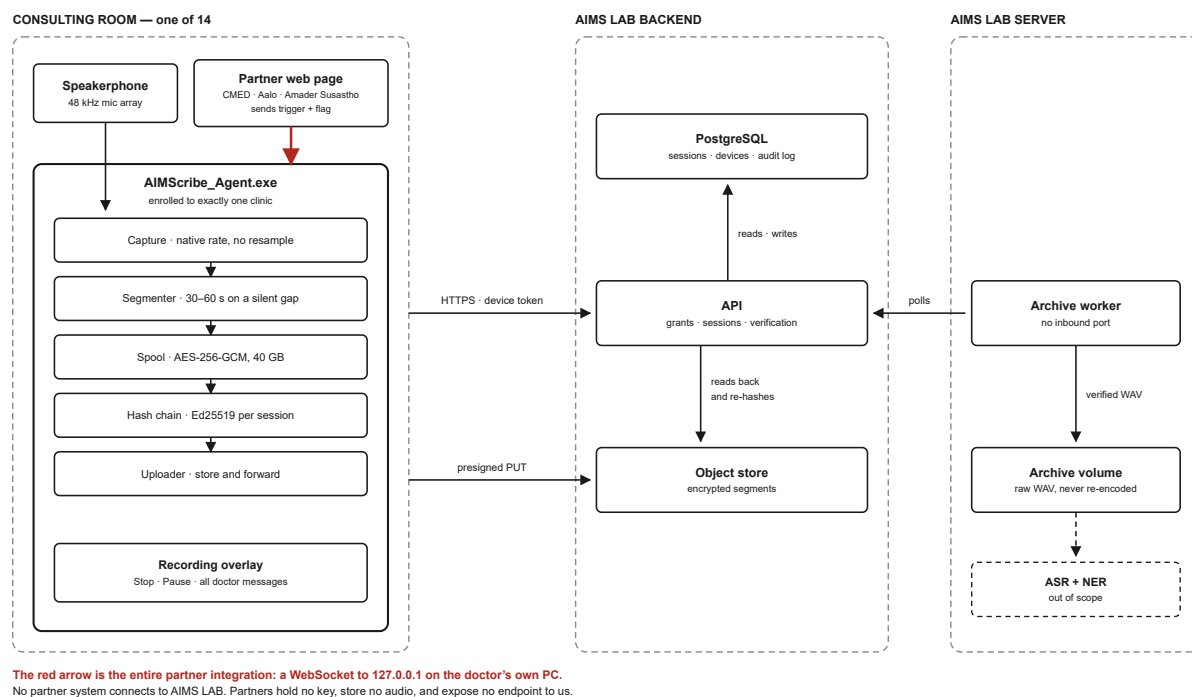


Figure 1. System architecture. A partner web application serves a page to the

browser on the consulting-room PC; that page reaches the AIMScribe agent only over a loopback WebSocket on the same machine. The agent captures, segments, encrypts, hash-chains and uploads audio to the AIMS LAB backend, which verifies every segment by reading it back from storage and re-hashing it. An archive worker inside AIMS LAB dials outward to collect verified sessions. No partner system ever connects to AIMS LAB infrastructure.

Note the direction of every arrow crossing an organisational boundary. **Nothing originates outside the clinic.** The archive worker dials out; it accepts no inbound connection. CMED is never called by us.

3.3 The pipeline, end to end

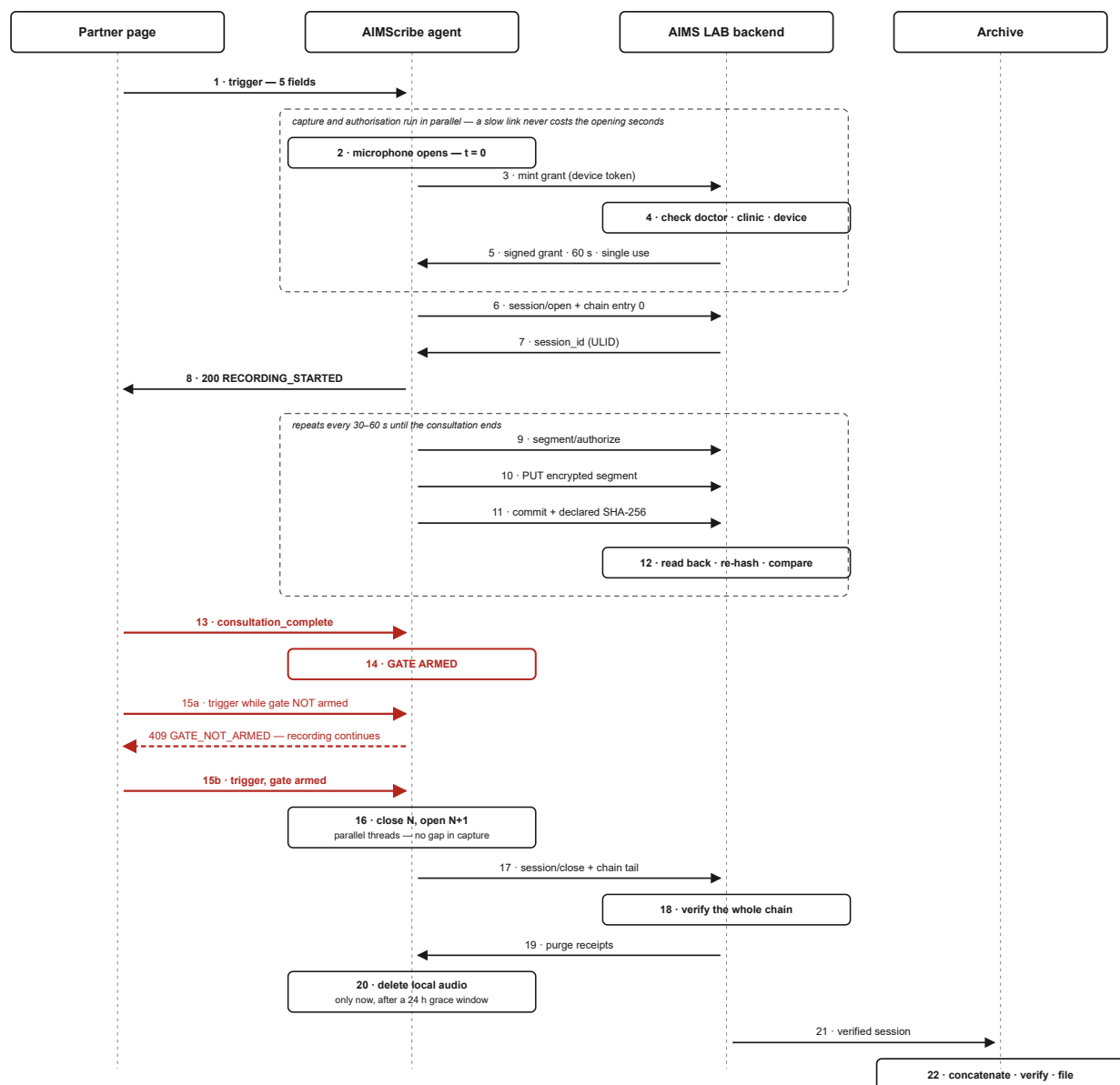


Figure 2. Life of a consultation. The microphone opens on the trigger while

authorisation proceeds in parallel, so a slow link never costs the opening seconds. Segments are sealed, uploaded and verified by server-side read-back throughout. A prescription-built flag arms the gate; until it arrives a stray trigger is refused (15a) and the recording continues undisturbed. Once armed, the next trigger closes one session and opens the next on parallel threads with no gap in capture. Local audio is deleted only after the chain verifies and purge receipts are issued.

This is the answer to *"the full AIMScribe pipeline from the recording to AIMScribe_Backend"*. Twenty steps, one consultation.

Read step 4 and step 11 together. Capture starts before authorisation finishes, and the two run in parallel. This is deliberate: a slow link must never cost the opening seconds of a consultation, which are the seconds in which the patient says why they came. If authorisation subsequently fails, the audio already captured is discarded under `SRS-GRT-08` — nothing is kept that was not authorised, but nothing is lost that was.

Read step 15 and step 16 together. The previous session closes and the next opens on separate threads, so the gap between two patients is not a gap in recording. This is **CON-08** made real.

3.4 Trust boundaries

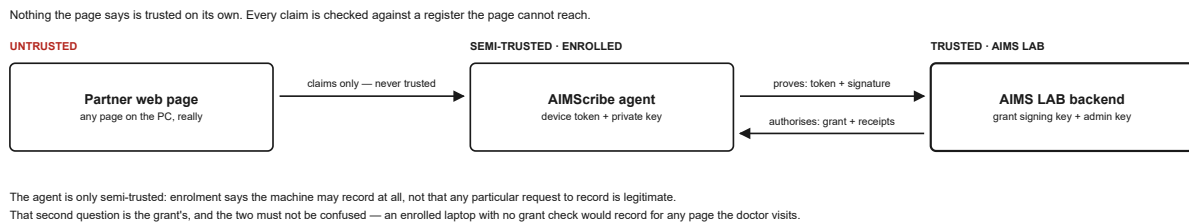


Figure 3. Trust boundaries. The partner page can only make claims; the agent proves its identity with a device token and an Ed25519 signature; the backend authorises with a signed grant and purge receipts. Nothing the page says is trusted on its own.

The single most important line in this diagram is the first arrow's label: **claims only**. Nothing the page sends is trusted on its own. Every claim is checked by the backend against a register the page cannot reach.

SRS-TOP-01 [M, I, AIMS] The agent shall bind its control API to loopback addresses only. It shall never listen on a routable interface.

SRS-TOP-02 [M, I, AIMS] The archive worker shall make outbound connections only. No inbound port shall be opened on the AIMS LAB server for the purposes of this system.

SRS-TOP-03 [M, I, Joint] No component shall require AIMS LAB to make an inbound or outbound connection to a CMED-operated service. **CON-02** .

Part II — The two questions CMED asks first

4. Device enrolment and machine identity

This section answers the question directly: how are enrolment tokens maintained today, and what will CMED have to build for them after integration?

Short answer: CMED builds nothing, and maintains nothing. Enrolment is a transaction between the consulting-room PC and the AIMS LAB backend. CMED is not a party to it, has no visibility into it, and is not affected by it. This was true before the integration was contemplated and remains true after it.

The reasoning in §4.6 is worth reading anyway, because it explains *why* the integration cannot weaken this, and why CMED's "we only read, we never write" constraint costs the design nothing.

4.1 What enrolment is for

Enrolment exists for exactly one purpose: **an unregistered laptop cannot run the software.** A PC that has not been enrolled by an AIMS LAB administrator will install, start, show its tray icon, and refuse to record.

It is worth being equally clear about what enrolment does *not* do, because the distinction determines who has to build what:

Enrolment does	Enrolment does not
Bind one PC to one clinic, permanently	Decide who the doctor is
Register the PC's Ed25519 public key with the backend	Authenticate a person
Issue the PC a long-lived bearer credential	Stop a random web page from triggering a recording
Give the backend a device register for revocation	Involve CMED in any way

The third row is the one that matters for §5. Enrolment answers *may this machine record at all*. It does not answer *is this particular request to record legitimate*. That second question is the grant's job, and the two must not be confused — an enrolled laptop with no grant check would happily record for any page the doctor happened to visit.

4.2 The credential hierarchy

Four distinct secrets exist. Confusing them is the most common cause of a misconfigured deployment, so they are tabulated explicitly.

#	Secret	Created by	Lives where	Lifetime	Scope
1	Enrolment token	AIMS LAB admin, via <code>POST /admin/enrollment-token</code>	An instruction sheet, then <code>%PROGRAMDATA%\AIMScribe\state\enrollment.token</code> , then deleted	72 h default, 720 h maximum, single use	Turns one into one enrolled device
2	Device token	Backend, at enrolment	<code>%PROGRAMDATA%\AIMScribe\state\device.token</code> , DPAPI-wrapped	Until revoked	Authentic every late backend c by that PC
3	Device private key	The agent, on first start, never leaves the PC	<code>%PROGRAMDATA%\AIMScribe\keys\</code> , DPAPI-wrapped	Until the machine is rebuilt	Signs eve chain ent
4	Grant signing key	AIMS LAB	Backend environment only	Rotatable	Signs recording authorisa (\$5)

SRS-ENR-01 [M, I, AIMS] The enrolment token shall be stored in the database as SHA-256 only. The plaintext shall exist nowhere but the generated instruction sheet, and shall be unrecoverable once that sheet is destroyed.

SRS-ENR-02 [M, T, AIMS] The enrolment token shall be single-use, enforced inside a single database transaction with row locking, so two concurrent presentations cannot both succeed.

SRS-ENR-03 [M, T, AIMS] The token shall carry a TTL between 1 and 720 hours, defaulting to 72.

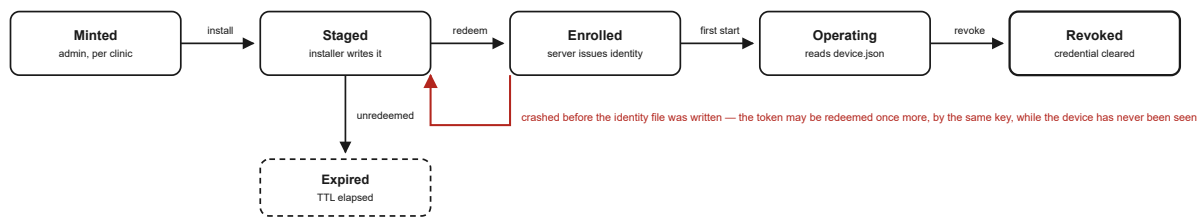
SRS-ENR-04 [M, I, AIMS] An unknown token, an expired token and an already-used token shall produce one identical error, so a caller learns nothing about which case occurred.

SRS-ENR-05 [M, I, AIMS] Both the device token and the device private key shall be wrapped by DPAPI at rest. Plaintext storage shall be permitted only under an explicit development flag that production configuration validation rejects.

SRS-ENR-06 [M, I, AIMS] Enrolment token sheets are credentials. They shall never be committed to a repository, and the generated `register.csv` shall deliberately contain no tokens.

4.3 The enrolment lifecycle

One token per laptop, consumed once. The machine receives its clinic; it never asserts one.



A backend that is merely unreachable does not burn the token: it stays on disk and the agent retries at the next start. Burning it would strand a PC in another district with a credential that cannot be reissued remotely.

Figure 4. Device enrolment lifecycle. A token is minted for a clinic, staged by the installer, redeemed at first start and replaced by a stored identity. An unreachable backend does not burn the token, and a crash before the identity is written returns the machine to the staged state.

The two self-transitions at the bottom are hard-won and deserve a note each.

A backend that is merely unreachable must not burn the administrator's token. If enrolment fails because the network is down, the token is deliberately left on disk and the agent retries at the next start. Burning it would strand a PC in another district with a credential that cannot be reissued remotely.

A token may be presented twice — but only in one narrow case. If the server committed the enrolment and the agent then failed to write its identity file (a full disk, a permissions problem, a crash between two writes), the machine has no credential and its token is spent. It can never recover on its own, and the only symptom is a PC reporting *not enrolled* while the server shows it enrolled perfectly well. That happened in the field. So a used token may be redeemed again by the *same* machine, proven by the *same* device public key, and only while the device it created has never once been seen — after the first heartbeat, a second presentation is a replay and is refused.

SRS-ENR-07 [M, T, AIMS] A failed enrolment caused by an unreachable backend shall leave the token on disk for retry.

SRS-ENR-08 [M, T, AIMS] A used token may be redeemed a second time only if all three hold: the device it created exists, that device has never been seen (`last_seen_at IS NULL`), and the presented public key matches the registered one. Any other repeat presentation shall be refused and logged as a replay attempt.

SRS-ENR-09 [M, T, AIMS] The agent shall write the device token before the identity file, so a crash between the two leaves the machine unenrolled rather than holding an identity it has no credential for.

SRS-ENR-10 [M, T, AIMS] The agent shall delete the enrolment token from disk immediately after successful redemption.

SRS-ENR-11 [M, T, AIMS] The agent shall compare the stored key fingerprint against the live device key at every start. On mismatch — a wiped key folder, a restored disk image — it shall refuse to record and require re-enrolment.

4.4 The administrator's procedure today

One token per **laptop**, not per doctor. A room with two laptops needs two.

1. The operator writes a **new, dated** CSV of machines to enrol. Existing fleet

files are never edited in place — a rewritten `laptops.csv` reads as *these hospitals were deleted*.

1. A tool calls `POST /api/v2/admin/enrollment-token` once per row, authenticated

by the admin key, and writes one instruction sheet per PC plus a token-free `register.csv`.

1. The administrator runs `AIMScribeSetup.exe` on the PC and pastes the token

into one of three fields. The other two are pre-filled.

1. The installer stages the token, registers a logon task and installs the pinned

public keys. The agent consumes the token on first start and deletes it.

1. The instruction-sheet folder is destroyed once the machines are live.

`doctor_id` in the enrolment record is **optional and decides nothing**. It labels the room in the paperwork. Most laptops are shared across shifts and have none — and if one is set, it is not used to attribute a consultation, because the doctor arrives from CMED with each trigger. This is `SRS-INV-03` in practice.

SRS-ENR-12 [M, I, AIMS] Fleet registration files shall be additive. A new dated file shall be written; an existing one shall not be rewritten.

SRS-ENR-13 [S, D, AIMS] Token minting shall be available to an operator through an interface that does not require editing the database directly.

SRS-ENR-14 [M, I, AIMS] `doctor_id` on an enrolment token shall be optional and shall never be used to attribute a recorded consultation.

4.5 Revocation

SRS-ENR-15 [M, T, AIMS] An administrator shall be able to revoke a device by identifier. Revocation shall clear the stored token hash, not merely set a flag, so a stolen laptop's credential stops working immediately.

SRS-ENR-16 [M, T, AIMS] A revoked device shall be refused at every authenticated route, and shall be told clearly enough that its tray reports the state to the user.

SRS-ENR-17 [M, I, AIMS] Every enrolment, re-issue and revocation shall be written to the append-only audit log with the acting administrator's name.

SRS-ENR-21 [M, T, AIMS] A device shall not be re-enrolled while segments remain unpurged in its spool. Re-enrolment issues a new device identity, and segments belonging to sessions opened under the previous identity cannot be committed with the new credential. The spool shall be drained to zero first, and the agent shall refuse re-enrolment until it is.

4.6 What changes for enrolment when CMED is integrated

Nothing. This is worth stating as a requirement rather than a reassurance, because the question will be asked again by CMED's security reviewer:

SRS-ENR-18 [M, I, Joint] Device enrolment shall require no participation from CMED. No CMED endpoint shall be called during enrolment; no CMED credential shall be presented; no enrolment state shall be stored in a CMED system; and no change to CMED software shall be required to enrol, re-enrol or revoke a device.

The reasoning, in full, because it is the part that generalises:

Enrolment binds a **machine** to a **clinic**. Both of those facts are AIMS LAB facts. AIMS LAB owns the laptop fleet, ships the laptops, assigns each one to a consulting room, and owns the archive whose top-level folder is that clinic's identifier. CMED knows none of this and has no reason to. A laptop moved from one clinic to another is an AIMS LAB logistics event, handled by re-enrolment, and CMED is not told because there is nothing for it to do.

Contrast that with the **doctor**, which is a CMED fact — CMED runs the rota and authenticates the people on it — and the **patient**, likewise. Those two arrive per consultation, in the trigger. The clinic does not, because it does not change.

This is also precisely why CMED's constraint — *we only read from our API, we never write* — costs this design nothing. Enrolment never needed a write from CMED. Neither, after the change in §5, does authorisation. The only thing CMED ever sends is a message to a WebSocket on the doctor's own PC, which is not a write to anything CMED owns.

The one operational touchpoint. When a new clinic comes online, AIMS LAB needs its `hospital_id` to be stable and needs to know which CMED clinic identifier corresponds to it. That is a one-row mapping agreed once per clinic, by email, and recorded on both sides. It is a fact, not an interface.

SRS-ENR-19 [M, I, Joint] For each clinic, CMED and AIMS LAB shall agree one stable mapping from CMED's clinic identifier to the AIMS LAB `hospital_id`, recorded in writing before the first recording at that site.

The register. Seven sites, two consulting rooms each, one laptop per room:

hospital_id	Site	Operator	Rooms	Status
HOSP001	Karail	Aalo	2	In service — signed recordings exist
HOSP002	Mirpur	Aalo	2	Assigned
HOSP003	Dholpur	Aalo	2	In service — signed recordings exist
HOSP004	Shyampur	Aalo	2	In service — signed recordings exist
HOSP005	Naryanganj	Aalo	2	To be assigned
HOSP006	Ershadnagar	Aalo	2	To be assigned
HOSP007	Amader Susastho	Amader Susastho	2	To be assigned

HOSP001 — HOSP006 are the six Aalo branches; HOSP007 is Amader Susastho. Fourteen rooms, fourteen laptops, two spares held centrally — sixteen enrolment tokens in total.

Identifiers HOSP001, HOSP003 and HOSP004 are **immutable in the strongest sense**: they appear inside signed chain payloads in the existing archive, so changing one would invalidate the Ed25519 signature over every entry that follows it. They cannot be renamed, and a signed payload cannot be edited by hand. Display names remain free to change at any time.

SRS-ENR-20 [M, T, AIMS] The backend shall reject a trigger whose clinic identifier does not map to the enrolled clinic of the device that presented it. See §5.5 for why this refuses rather than warns.

5. Authorisation: the recording grant

5.1 The problem enrolment leaves open

An enrolled laptop may record. That is not the same as *this request to record is legitimate*.

Consider the gap concretely. A doctor's PC is enrolled and working correctly. The doctor opens a browser tab to an unrelated website. That page runs JavaScript. It opens a WebSocket to `127.0.0.1:5050` — which any page on that machine can attempt — and sends `start`. Without a second control, the agent records.

That is the gap the grant closes, and it is the reason the grant exists as a separate mechanism rather than being folded into enrolment.

5.2 What a grant is

A short-lived, single-use, Ed25519-signed statement that a named doctor may record a named patient at a named clinic.

Property	Value	Why
Algorithm	EdDSA / Ed25519	Small, fast, no parameter choices to get wrong
Lifetime	60 s	Long enough for a click to reach the agent; short enough that a captured grant is worthless before it can be replayed
Clock leeway	5 s	Consulting-room PCs drift
Replay protection	Single-use <code>jti</code> , tracked by the agent	A captured grant is refused even inside its 60 s
Audience	<code>aimscribe-recorder</code>	A grant cannot be repurposed for another service
Required claims	<code>consent_obtained</code> , <code>patient_ref</code> , <code>doctor_id</code> , <code>hospital_id</code>	Consent is a precondition, not an afterthought

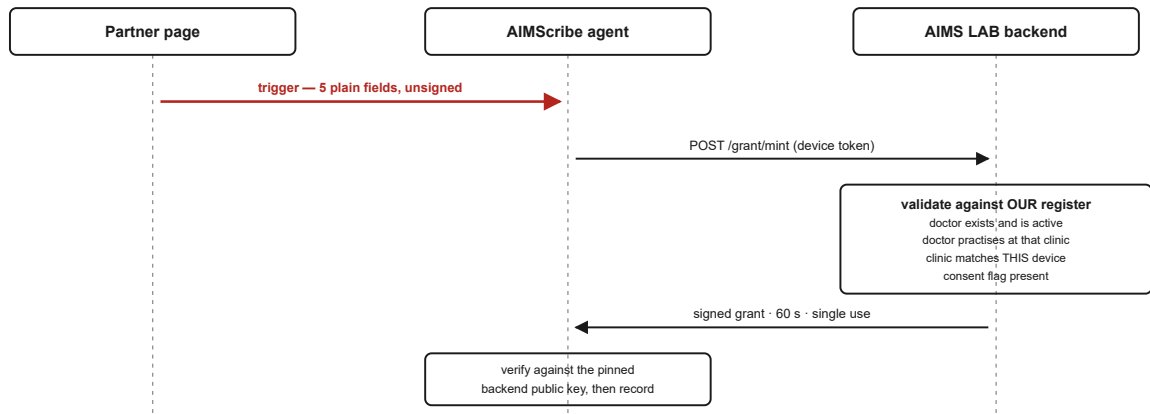
5.3 Where grants are minted — the change that matters to CMED

Today, in the development environment, the dummy CMED application we wrote mints the grant itself and holds a private key.

Real CMED will not do that, will not hold a key, and has said so plainly. That constraint is entirely acceptable, and the design that replaces it is *stronger* than the one it replaces.

Target: grant minting moves to the AIMS LAB backend.

The change that matters most to a partner: you hold no key, sign nothing, and expose no endpoint.



Before, the agent trusted what a page claimed. Now the backend checks every claim against a register the page cannot reach.

Figure 5. Grant minting moves to the AIMS LAB backend. The partner page sends five plain unsigned fields; the backend validates them against its own register and returns a short-lived single-use grant. The partner holds no key and signs nothing.

Compare the two honestly. Before, the agent trusted what a page claimed, because the page held the key that made the claim believable. Now the backend checks every claim against a register the page cannot reach, and the page holds nothing.

	Old (dummy)	New (real CMED)
Who holds a private key	The web app	AIMS LAB backend only
What the page sends	A signed token it minted	Five plain fields
Who validates the doctor	Nobody	The backend, against its register
Who validates the clinic	Nobody	The backend, against the device's enrolment
CMED's key-management burden	Real	None

SRS-GRT-01 [M, I, AIMS] Grant minting shall be performed exclusively by the AIMS LAB backend. No grant signing key shall be issued to, held by, or required of CMED.

SRS-GRT-02 [M, T, AIMS] The mint endpoint shall be authenticated by the calling device's token, and shall mint only for that device's enrolled clinic.

SRS-GRT-03 [M, T, AIMS] The backend shall validate, before minting: the doctor exists and is active; the doctor is associated with that clinic; the clinic matches the device's enrolment; and consent is asserted. Failure of any check shall refuse the mint with a distinguishable code.

SRS-GRT-04 [M, T, AIMS] The agent shall verify every grant against a public key pinned at installation, checking signature, issuer, audience and expiry with no more than 5 s leeway.

SRS-GRT-05 [M, T, AIMS] The agent shall enforce single use by `jti` and refuse a repeat within the grant's lifetime.

SRS-GRT-06 [M, T, AIMS] The agent shall refuse to start a recording when no grant verification key is installed. It shall not fall back to unauthorised recording under any configuration reachable in production.

SRS-GRT-07 [M, D, AIMS] The mint round-trip shall not delay the opening of the microphone. Capture shall begin on receipt of the trigger and authorisation shall proceed in parallel.

SRS-GRT-08 [M, T, AIMS] If authorisation fails after capture has begun, audio captured under that trigger shall be discarded and not written to the spool.

5.4 What CMED must build for authorisation

Nothing cryptographic. Concretely, CMED's authorisation-related work is:

1. Send five fields instead of a signed token. (Fewer moving parts than the dummy.)
2. Read the reply and branch on `code`.

That is the entire list. There is no key to generate, store, protect, rotate or audit; no signing library to add; no clock synchronisation to manage; and no secret in CMED's environment variables.

SRS-GRT-09 [M, I, CMED] CMED shall send the trigger as plain JSON fields and shall not be required to compute any signature, hash or MAC.

5.5 Clinic mismatch refuses rather than warns

If a trigger names a clinic that does not match the device's enrolment, one of two things is true: the clinic mapping is wrong, or a laptop is physically in a different building from where it is registered. Both produce mislabelled evidence, and mislabelled evidence is worse than no evidence, because it is believed.

SRS-GRT-10 [M, T, AIMS] A clinic mismatch shall refuse the recording with `401 CLINIC_MISMATCH` and raise an operational alert. It shall not record and warn.

Against the dummy, recording-and-warning was the right behaviour. Against real CMED it is not. **OD-03** offers CMED the option of keeping warn-and-record for the pilot only.

Part III — Requirements

6. External interface requirements

6.1 Interface 1 — CMED page → agent

This is the only interface CMED implements. Everything in this subsection is CMED-facing; everything after it is AIMS LAB's own obligation.

6.1.1 Transport

Property	Value
Endpoint	<code>ws://127.0.0.1:5050/ws</code>
Health probe	<code>GET http://127.0.0.1:5050/health</code> — unauthenticated, no side effects
Encoding	UTF-8 JSON, one object per frame
Maximum frame	64 KB; larger frames are refused
Reconnection	Exponential backoff, 1 s → 30 s, indefinitely
Origin header	Mandatory; must exactly match an allowlisted value

Why plain `ws://` from an HTTPS page is permitted. The W3C *Secure Contexts* specification treats loopback as a potentially trustworthy origin, so browsers do not apply mixed-content blocking to `ws://127.0.0.1`. The alternative — a TLS certificate for `127.0.0.1` on thirty machines — is worse in every dimension: certificates expire, and an expired certificate on a clinical PC stops recordings silently at 3 a.m.

Why WebSocket rather than HTTP. Three reasons, in order of importance. The agent must *push* state to the page — a stopped recording, a full disk, a lost microphone — and HTTP cannot push. WebSockets are not subject to CORS, so the agent enforces its own origin check rather than relying on a browser mechanism that does not apply. And a persistent connection means the page knows the agent is alive without polling. There is deliberately **no** `POST /session/start` route; starting a recording requires the WebSocket, because that is where grant verification and origin pinning live, and a second start path would be a weaker one.

SRS-IF1-01 [M, D, CMED] CMED shall connect to `ws://127.0.0.1:5050/ws` from the doctor's browser and maintain the connection for the session.

SRS-IF1-02 [M, T, CMED] CMED shall reconnect with exponential backoff on disconnection and shall not busy-loop.

SRS-IF1-03 [M, I, CMED] If CMED enforces a Content-Security-Policy, its `connect-src` directive shall permit `ws://127.0.0.1:5050`. Without this the browser blocks the connection before the agent sees it.

SRS-IF1-04 [S, D, CMED] CMED should probe `GET /health` before the first connection attempt to distinguish *agent absent* from *agent failing*.

6.1.2 Admission control

Three checks run in order **before** the handshake completes. Each closes the socket with code `4403`.

#	Check	Rule	Defends against
1	Origin	Must appear exactly in the allowlist. Absent or literal <code>null</code> is refused.	A random page on the PC opening a socket
2	Host	Must be an expected loopback authority	DNS rebinding — <code>http://evil.example</code> can resolve to <code>127.0.0.1</code> , and only <code>Host</code> distinguishes it
3	Peer address	Must be <code>127.0.0.1</code> or <code>::1</code>	A non-loopback client

Check 3 alone is not sufficient and was the v1 flaw: the peer address of *any* browser on the PC is loopback, so that check always passed.

SRS-IF1-05 [M, I, CMED] CMED shall provide AIMS LAB with the exact origins — scheme, host and port — from which the page will connect, for production and for staging. Wildcards shall not be accepted.

SRS-IF1-06 [M, T, AIMS] The agent shall refuse a connection whose origin is absent, `null`, or not exactly allowlisted, and shall refuse a startup configuration containing a wildcard origin.

6.1.3 Signal 1 — the consultation trigger

Sent when the doctor opens a patient's details or history.

```
{
  "command": "start",
  "request_id": "cmed-7f3a2b91",
  "trigger": {
    "patient_id": "P0012345",
    "doctor_id": "DR0042",
    "clinic_id": "CMED-DHK-BANANI-01",
    "start_time": "2026-08-25T10:14:32+06:00",
    "date": "2026-08-25",
    "consent_obtained": true,
    "consent_method": "verbal_at_reception",
    "patient_name": "optional, display only"
  }
}
```

Field	M/S/C	Type	Rule
patient_id	M	string	<code>^[A-Za-z0-9_-]{1,64}\$</code> . Stable for the patient.
doctor_id	M	string	Same pattern. No default, ever (SRS-INV-03).
clinic_id	M	string	Same pattern. Must map to the device's enrolled clinic.
start_time	M	string	RFC 3339 with explicit offset
date	M	string	<code>YYYY-MM-DD</code> , local clinic date
consent_obtained	M	boolean	Must be <code>true</code> . <code>false</code> or absent refuses.
consent_method	S	string	Free text ≤ 64 chars, for the audit record
patient_name	C	string	Display only, ≤ 120 chars, never stored
request_id	S	string	Echoed in the reply for correlation

6.1.4 The reply — the "200 OK" contract

Every command receives exactly one reply, carrying a numeric `status` mirroring HTTP semantics and a stable string `code` .

```
{
  "event": "ack",
  "command": "start",
  "request_id": "cmed-7f3a2b91",
  "status": 200,
  "code": "RECORDING_STARTED",
  "message": "Recording started.",
  "data": {
    "session_id": "01JB8XQ4M7YZ2K9V3N5P6R8T0W",
    "started_at": "2026-08-25T10:14:32+06:00",
    "armed": false,
    "supersedes": "01JB8W2H..."
  }
}
```

SRS-IF1-07 [M, T, AIMS] Every command shall receive exactly one reply carrying `status` , `code` and, where supplied, the caller's `request_id` .

SRS-IF1-08 [M, D, AIMS] A successfully started recording shall reply `status: 200` , `code: "RECORDING_STARTED"` , including the `session_id` .

SRS-IF1-09 [M, I, CMED] CMED shall branch on `code` , never on `message` . Message text is shown to doctors and will be reworded without notice.

SRS-IF1-10 [M, T, AIMS] The full response-code set of Appendix A shall be implemented, and no code outside it shall be returned for these commands.

6.1.5 Signal 2 — the prescription-built flag

Sent when the doctor presses **Build Prescription** — the action that converts the raw tabular entry into a proper prescription. This action is performed for every patient, because the paramedic in the investigation chamber reads the result, so it is a reliable signal that the consultation has reached its conclusion.

```
{
  "command": "consultation_complete",
  "request_id": "cmed-7f3a2b92",
  "patient_id": "P0012345",
  "occurred_at": "2026-08-25T10:26:11+06:00"
}
```

What this signal does and does not do. It does **not** stop the recording. The doctor builds the prescription, prints it, hands it over, and then counsels the patient for another one to two minutes — and that counselling is clinically valuable audio. The flag *arms the gate*: it permits the **next** trigger to end this session. Recording continues until the next patient arrives, typically 20–30 seconds later.

SRS-IF1-11 [M, D, CMED] CMED shall send `consultation_complete` when the prescription-build action succeeds, carrying the same `patient_id` as the trigger.

SRS-IF1-12 [M, T, AIMS] The flag shall not stop, pause or otherwise alter the current recording. It shall only set the gate to armed.

SRS-IF1-13 [M, T, AIMS] A flag naming a patient other than the open session's shall be refused `PATIENT_MISMATCH` and shall not arm the gate.

SRS-IF1-14 [M, T, AIMS] A repeated flag for the already-armed session shall succeed idempotently with `GATE_ALREADY_ARMED`.

SRS-IF1-15 [M, I, AIMS] The flag's value shall be used only as an internal safety interlock. Nothing from it shall be stored as clinical content.

6.1.6 Events the agent pushes

Event	When	What CMED should do
<code>status</code>	On connect, and on every state change	Update any indicator
<code>session_started</code>	Capture is running	Optional confirmation
<code>session_stopped</code>	Session closed, with reason	Update the indicator
<code>session_paused</code> / <code>session_resumed</code>	Doctor used the overlay	Update the indicator
<code>gate_armed</code>	Flag accepted	Optional
<code>warning</code>	Disk low, microphone lost, backend unreachable	Optionally surface to the doctor

SRS-IF1-16 [M, I, AIMS] **CMED shall not be required to render anything.** Every message intended for the doctor — recording state, gate state, level prompts, disk and microphone warnings, errors — shall be delivered by the AIMSScribe overlay (§7.8), which AIMS LAB builds and controls.

SRS-IF1-18 [C, D, CMED] CMED *may* additionally show its own indicator from the pushed events, but nothing in this specification depends on it and the system is fully usable when CMED renders nothing at all.

This is a deliberate consequence of the constraint that CMED will not modify its software beyond sending two signals. If doctor-facing messaging depended on CMED's user interface, every future message would require a CMED release. It does not: the overlay is ours, so we can change what a doctor is told without asking anyone.

6.1.7 When the agent is absent

This is the requirement that outranks the rest of the document.

SRS-IF1-17 [M, T, CMED] A failure of any kind in the AIMSScribe integration — connection refused, timeout, error reply, malformed frame — shall be caught, logged, and otherwise ignored by the CMED page. It shall never raise a dialog, block navigation, or prevent the doctor from seeing the patient.

A recording that fails is a recording lost. A clinical system that stops because a recorder failed is a patient not seen. These are not comparable, and the design never treats them as such. Appendix B shows this as three characters of JavaScript: `.catch(() => {})`.

6.2 Interface 2 — agent → backend (AIMS LAB internal)

Documented so CMED's reviewers can see the whole chain of custody, not because CMED implements any of it.

Route	Auth	Purpose
POST /device/enroll	Enrolment token	Exchange a one-time token for an identity token
POST /grant/mint	Device token	Authorise one consultation (§5, to be built)
POST /session/open	Device token	Open a session, submit chain entry 0
POST /segment/authorize	Device token	Obtain a 300 s presigned PUT for one segment
POST /segment/commit	Device token	Declare a segment uploaded, with its SHA-256
POST /session/pause · /resume	Device token	Record a supervised pause with its reason
POST /session/close	Device token	Close and submit the chain tail
GET /session/{id}/receipts	Device token	Collect purge receipts
POST /heartbeat	Device token	Liveness, spool depth, version
GET /doctors	Device token	Typing suggestions for this clinic only
GET /archive/pending · POST /archive/complete	Worker key	Archive worker
/admin/*	Admin key	Clinics, doctors, tokens, revocation, alerts

SRS-IF2-01 [M, I, AIMS] All agent-backend traffic shall use TLS 1.2 or later.

SRS-IF2-02 [M, T, AIMS] Three separate bearer credentials shall exist — device, worker, admin — and shall not be interchangeable.

SRS-IF2-03 [M, T, AIMS] A device shall be able to read only its own clinic's data. `GET /doctors` for another clinic shall return 403.

6.3 Interface 3 — backend → archive worker

SRS-IF3-01 [M, I, AIMS] The archive worker shall poll outbound and shall never accept an inbound connection.

SRS-IF3-02 [M, T, AIMS] The worker shall claim work in batches with a visibility guard so two workers cannot archive the same session twice.

7. Functional requirements

7.1 Audio capture — CAP

ID	Requirement	Pri	Ver	Owner
SRS-CAP-01	Capture shall be WAV PCM, 44.1 kHz, mono, 16-bit — 88,200 bytes/second.	M	I	AIMS
SRS-CAP-02	Audio shall never be re-encoded to a lossy format at any point before the archive.	M	I	AIMS
SRS-CAP-03	Capture shall begin within 500 ms of an accepted trigger.	M	A	AIMS
SRS-CAP-04	Capture shall run on a dedicated thread and shall not be blocked by network, disk or UI work.	M	I	AIMS
SRS-CAP-05	Loss of the input device mid-session shall raise a warning event, attempt recovery, and preserve everything captured so far.	M	T	AIMS
SRS-CAP-06	The agent shall detect a persistently silent session — RMS below ~-52 dBFS across two consecutive segments — and latch one <code>silent_session</code> notification per session.	S	T	AIMS

On SRS-CAP-02 . Opus at 32 kbps would cut storage by an order of magnitude and was rejected. The archive is evidence. A lossy re-encode cannot be hashed back to what the microphone heard, which destroys the property the entire chain exists to establish. **CON-07** .

On SRS-CAP-06 . This is the "floor noise" notification. It is a *level* check on completed segments — a microphone muted at the hardware switch, unplugged, or pointed at the wrong input. It is distinct from, and must not be confused with, the segmenter's adaptive noise floor in **SRS-SEG-03** , which is a *boundary-finding* mechanism and raises no notification at all.

7.1a The capture path — CAP continued

Between the microphone capsule and the WAV file sit several stages of optional processing, and every one of them is a place where a quiet talker disappears. A consultation has two talkers at very different levels: the doctor is close to the device and speaks with professional projection; the patient is further away, often unwell, often elderly, and speaks quietly. Any processing tuned to "isolate the dominant near-field talker" — which is what conferencing hardware is built to do — will remove the patient.

Measurements on 128 archived research sessions established the failure signature. In degraded sessions the noise floor sits at -87 to -97 dBFS with **8.7% of frames at literal digital zero**, and the quiet talker lands at -56 to -69 dBFS. In healthy sessions from the same fleet the floor is -60 to -71 dBFS and the quiet talker is at -36 to -42 dBFS. The loud talker is at -18 to -21 dBFS in both. A microphone in a room with a fan cannot produce digital zero; only a gate can.

ID	Requirement	Pri	Ver	Owner
SRS-CAP-07	Capture shall use the device's native sample rate and channel count , with no operating-system resampling or downmix in the path.	M	T	AIMS
SRS-CAP-08	Capture shall use a low-latency host API that bypasses the shared audio engine where one is available (WASAPI on Windows). The legacy MME path shall not be used.	M	I	AIMS
SRS-CAP-09	The capture device shall be selected by name and host API , never by numeric index, because indices move between reboots and USB ports.	M	T	AIMS
SRS-CAP-10	Device-side or OS-side automatic gain control, noise suppression and gating shall be disabled wherever the platform exposes them.	M	I	AIMS
SRS-CAP-11	A machine shall not be accepted into service until a commissioning measurement shows < 0.5% of frames at digital zero over 60 s of occupied-room audio.	M	T	AIMS
SRS-CAP-12	Commissioning shall record and store, per room: peak level, noise floor (P10), quiet-talker level (P25 of speech frames) and loud-talker level (P90).	M	T	AIMS
SRS-CAP-13	Where the platform cannot expose the device's processing, hardware without onboard processing shall be used instead. A conferencing device whose DSP cannot be disabled is not fit for archival capture.	M	A	AIMS

Acceptance envelope, derived from the fleet's own healthy population rather than chosen a priori:

Quantity	Target	Reject
Peak (loud talker)	−6 to −12 dBFS	above −3 dBFS (clipping)
Noise floor (P10)	−60 to −75 dBFS	below −85 dBFS (gating)
Quiet talker (P25 of speech)	above −45 dBFS	below −50 dBFS
Frames at digital zero	< 0.5%	≥ 0.5%

Gain is not the lever. The fleet already records the doctor at −18 to −21 dBFS, which is correct, and at least one archived session clipped at 0.0 dBFS. Raising input gain moves the loud talker into clipping without lifting the quiet talker at all, because the quiet talker is not attenuated — it is being gated. Physical placement is worth 6–10 dB and costs nothing: sound falls 6 dB per doubling of distance, so moving the device toward the patient transfers level from the talker who has 20 dB to spare to the one who needs it.

7.1b Speech-level monitoring — LVL

The condition that degrades transcription is the same condition that breaks segmentation, so one measurement serves both. It must be expressed as a **ratio**, never an absolute level: a loud room with a loud talker is fine.

ID	Requirement	Pri	Ver	Owner
SRS-LVL-01	The agent shall compute, per sealed segment, the noise floor and the 25th and 90th percentiles of speech-frame level.	M	T	AIMS
SRS-LVL-02	These figures shall be reported to the backend as operational metrics for every session.	M	T	AIMS
SRS-LVL-03	When the quiet talker falls below -50 dBFS while the loud talker is above -25 dBFS, the overlay shall show a non-modal prompt to ask the patient to speak a little louder.	S	D	AIMS
SRS-LVL-04	When both talkers are low, the prompt shall instead report a microphone problem, because that is not something a patient can fix.	S	T	AIMS
SRS-LVL-05	When the loud talker exceeds -3 dBFS the agent shall warn that the recording is clipping.	S	T	AIMS
SRS-LVL-06	Level prompts shall latch once per session , shall never be modal, and shall never require dismissal.	M	T	AIMS
SRS-LVL-07	The first segment of a session shall be exempt, while the noise-floor estimate is still converging.	M	T	AIMS
SRS-LVL-08	Rooms chronically below the envelope shall be reported to operations as an environment or hardware defect , not addressed by prompting the doctor.	M	I	AIMS

SRS-LVL-08 matters more than the prompt does. A room that sits below the envelope every day needs a microphone moved or replaced; asking a clinician to raise their voice thirty times a day is not a fix, and treating it as one hides a defect behind a human.

7.2 Segmentation — SEG

ID	Requirement	Pri	Ver	Owner
SRS-SEG-01	Segments shall be 30–60 s, closing at a silent point inside the window.	M	T	AIMS
SRS-SEG-02	If no silence is found by 60 s, the segmenter shall continue up to a 15 s grace, then force a cut at 75 s.	M	T	AIMS
SRS-SEG-03	Silence detection shall use linear RMS with a 320 threshold and a 3.0 s hold, against an adaptive noise floor, combined with zero-crossing rate.	M	I	AIMS
SRS-SEG-04	Segmentation shall run on its own thread, so a slow disk cannot stall capture.	M	I	AIMS
SRS-SEG-05	Segments shall be numbered contiguously from zero within a session, with no gaps.	M	T	AIMS
SRS-SEG-06	Silence detection shall be level-independent : the speech estimate shall be seeded from the opening seconds of audio rather than from a fixed constant, so a quiet or gated input degrades gracefully instead of classifying an entire clip as silence.	M	T	AIMS
SRS-SEG-07	At least 80% of segments in a commissioned room shall fall within the 30–60 s target window, measured over a clinic day.	M	A	AIMS

Why 30–60 s and not three minutes. A segment is the unit of upload, retry and loss. At three minutes a single failure put three minutes of consultation at risk and held everything behind it. At a minute, exposure is a minute, transcripts begin arriving while the doctor is still talking, and a poor

connection recovers in smaller steps. The server merges the clips back into one file, so nothing is lost by cutting often.

7.3 Local durability — SPL

ID	Requirement	Pri	Ver	Owner
SRS-SPL-01	Every sealed segment shall be encrypted AES-256-GCM with a per-segment key, before it touches disk.	M	I	AIMS
SRS-SPL-02	The spool master key shall be DPAPI-wrapped at rest.	M	I	AIMS
SRS-SPL-03	Segment state shall follow PENDING → COMMITTED → RECEIPTED → PURGED, with QUARANTINED as a terminal branch.	M	T	AIMS
SRS-SPL-04	State transitions shall be recorded in an fsynced append-only <code>journal.jsonl</code> before the corresponding action.	M	T	AIMS
SRS-SPL-05	The spool shall survive an abrupt power loss with no partially-written segment accepted as complete.	M	T	AIMS
SRS-SPL-06	The spool shall hold 40 GB by default — about 135 recording-hours, roughly three weeks of one room's work.	M	A	AIMS
SRS-SPL-07	The agent shall warn at 50 % and alarm at 80 % of the spool cap, and shall refuse to start a session below 20 GB free.	M	T	AIMS
SRS-SPL-08	Local audio shall be deleted only on a valid purge receipt, after a 24 h grace period.	M	T	AIMS
SRS-SPL-09	A QUARANTINED segment shall be delivered to the server before deletion, exactly as a verified one is. It shall not be deleted while undelivered, and shall not be retained once delivered.	M	T	AIMS
SRS-SPL-13	No segment shall remain on the workstation beyond the drainage interval once connectivity permits delivery. The age of the oldest undelivered segment shall be reported in the heartbeat.	M	T	AIMS
SRS-SPL-14	The spool shall reach zero whenever the backend has been reachable and idle for one drainage cycle. A non-empty spool on a connected machine is a defect and shall alert.	M	T	AIMS

Why the thresholds in SRS-SPL-07 exist at all. The spool does not fill because audio accumulates in normal operation — in normal operation it drains continuously and each segment is deleted within a day. It fills only when deletion is *blocked*, and deletion is blocked whenever a purge receipt cannot be obtained: the clinic's connectivity is down, the backend is unreachable, the archive worker has stopped issuing receipts, or the session is quarantined. The 40 GB is a buffer sized for three weeks of that condition, and 50 %, 80 % and the 20 GB floor are the graduated warnings that the buffer is filling. They are not a sign of a design fault; they are the instrument that reports one.

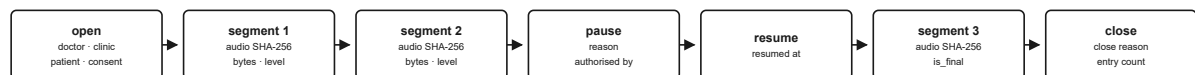
SRS-SPL-13 and SRS-SPL-14 state the consequence of §7.5a.2: because every segment is delivered whatever its verification outcome, a connected machine has no reason to hold anything. A spool that is not empty on a machine that has been online is therefore not a capacity question but a fault, and is reported as one.

The volatile-memory exposure, stated honestly. Between the capture buffer and the pre-seal queue, approximately **99 seconds** of audio exists only in RAM — a 75 s buffer plus a 512×4096 -byte queue of about 23.8 s. A hard power loss during that window loses that audio. Segment sealing bounds this; it does not eliminate it.

ID	Requirement	Pri	Ver	Owner
SRS-SPL-10	An intermediate durability checkpoint shall be added beneath per-segment sealing, reducing volatile exposure without altering the segmentation, chain or encryption design.	S	T	AIMS
SRS-SPL-11	The checkpoint interval shall be set from measured fsync latency on the deployed clinical hardware, not chosen a priori.	S	A	AIMS
SRS-SPL-12	The checkpoint shall be validated across repeated induced-termination trials before being described as an operational capability.	S	T	AIMS

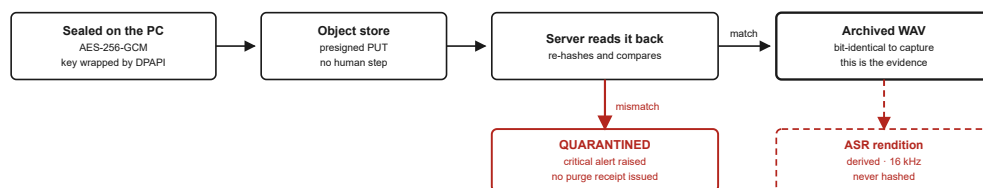
7.4 Integrity — CHN

A · THE SESSION HASH CHAIN — one per consultation, signed by the device key



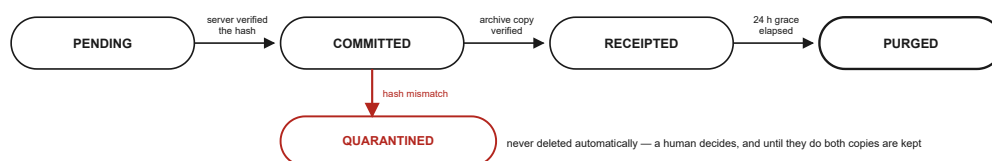
Every entry embeds the previous entry's digest and carries its own signature, so a removed, reordered or edited entry is detectable. The backend verifies the complete chain at close.

B · WHAT HAPPENS TO THE AUDIO — uploaded automatically, deleted only by proof



The upload is automatic and needs no human. Verification happens on arrival, not before it — so a bad segment reaches storage, and is then caught there. What a mismatch withholds is the purge receipt. Both copies survive, the session is never archived, and an operator decides what happens next.

C · WHEN LOCAL AUDIO MAY BE DELETED



A purge receipt is the only authority to delete audio.

Nothing on the consulting-room PC is removed until the server has proved, by re-hashing the stored bytes, that an archived copy exists and matches. A quarantined session therefore holds its local audio indefinitely. That is deliberate: the disk fills before the evidence disappears.

Figure 6. Chain of custody. Each session carries an Ed25519-signed hash chain

whose entries link by the previous entry's digest, so omission, reordering or edit is detectable. A sealed segment is encrypted on the PC, uploaded, then read back and re-hashed by the server before acceptance. The archived WAV is bit-identical to what was captured and is the hashed evidence; the speech-recognition rendition is derived from it and never hashed. Local audio moves PENDING → COMMITTED → RECEIPTED → PURGED, and a hash mismatch quarantines the session so nothing is deleted automatically.

ID	Requirement	Pri	Ver	Owner
SRS-CHN-01	Each session shall carry an Ed25519-signed hash chain with entries of type <code>open</code> , <code>segment</code> , <code>pause</code> , <code>resume</code> , <code>close</code> .	M	T	AIMS
SRS-CHN-02	Each entry shall include the previous entry's digest, making omission and reordering detectable.	M	T	AIMS
SRS-CHN-03	Hashing shall be domain-separated SHA-256 — <code>aimscribe.chain.v2</code> , <code>aimscribe.receipt.v2</code> , and equivalents for spool and audit.	M	I	AIMS
SRS-CHN-04	Entries shall be signed by the device key registered at enrolment.	M	T	AIMS
SRS-CHN-05	The backend shall verify the complete chain at session close, and quarantine the session on any failure.	M	T	AIMS
SRS-CHN-06	Chain entries shall be serialised as canonical JSON so a signature is reproducible byte for byte.	M	T	AIMS
SRS-CHN-07	A broken chain shall be repaired only by a fix in system code. Manual editing of chain data is prohibited.	M	I	AIMS

On SRS-CHN-07 . A chain repaired by hand proves nothing. If a chain breaks, the defect is in the code that produced it, and hand-patching the evidence destroys the only reason the evidence exists.

7.5 Upload and server-side verification — UPL

ID	Requirement	Pri	Ver	Owner
SRS-UPL-01	Segment upload shall use a presigned PUT valid for 300 s, obtained per segment.	M	T	AIMS
SRS-UPL-02	On commit, the backend shall read the object back from storage and re-hash it, comparing in constant time against the claimed SHA-256.	M	T	AIMS
SRS-UPL-03	A hash mismatch on arrival shall quarantine the session and alert.	M	T	AIMS
SRS-UPL-04	Object keys shall be built from the session ULID and a prefix, never from the patient reference.	M	I	AIMS
SRS-UPL-05	Upload shall retry with backoff and shall never block capture or segmentation.	M	T	AIMS
SRS-UPL-06	The agent shall function fully offline, spooling locally and uploading when connectivity returns.	M	D	AIMS

On SRS-UPL-02 — and it has a cost worth naming. Because every segment is read back and re-hashed on the API server, the API tier carries the full audio volume in *both* directions. This is not a thin control plane, and §10 sizes it accordingly. The alternative — trusting the client's hash — would make the whole chain decorative, since a corrupted upload would be recorded as verified.

On SRS-UPL-04 . Object keys leak into access logs, metrics dashboards and error traces. A patient reference in a key is a patient reference in all three.

7.5a Recovering audio that failed verification — REC

The gap this closes. Today a hash mismatch stops the session dead. The uploader sees a quarantined segment, logs, emits one alert and `continue s` — for that session, permanently. The audio stays sealed on the consulting-room PC and **never reaches the server at all**. On restart the session is re-adopted from its journal, reaches the same check, and stops again.

Stopping is right; stopping *forever* is not. The agent cannot currently tell the difference between two very different failures:

Cause	Local original	Correct response
Corruption in transit	intact	re-upload — the bytes on disk are good
Corruption in object storage	intact	re-upload to a fresh key
Local file decayed on disk	damaged	do not retry; escalate, keep, and mark it

Local damage is already caught, and caught early. `SessionSpool.read_segment` decrypts and re-verifies the segment against its recorded hash *before every upload attempt*, and a failure quarantines it locally as `local_segment_unreadable` without ever sending it. So by the time a segment reaches the server, its bytes were verified good microseconds earlier.

That has a sharp consequence: **a server-side hash mismatch is almost never damaged audio**. The local original was intact when it left. What is missing is not another local check but a retry after a server rejection — the agent quarantines and stops instead of re-reading and trying again.

And most quarantines are not about the audio at all. Of the five triggers in §7.5a.1 below, only one concerns the audio bytes; the rest are chain ordering and continuity failures, where every second of the recording is perfectly intact. A dropped pause entry once quarantined a real consultation whose audio was flawless.

7.5a.1 Every way a session can be quarantined

#	Trigger	Where	Is the audio damaged?	Likely cause
1	local_segment_unreadable	agent, before upload	Yes, possibly	Disk bit-rot over the 3-week spool window; truncated write after power loss; filesystem damage; a bit flipped in non-ECC RAM between encrypt and write; antivirus altering the file; a DPAPI key that no longer decrypts after a profile rebuild
2	segment hash mismatch on arrival	backend, at commit	No — it was verified on read	Upload truncated by a dropped connection; presigned URL expiring mid-PUT; a retried PUT leaving a partial object; a fault on the storage side
3	chain entry rejected	backend, at commit	No	An entry arriving before its predecessor; a dropped pause or resume breaking prev_hash continuity; a device key that no longer matches
4	duplicate seq_no with a different hash	backend, at commit	No	A commit that succeeded but whose response was lost, retried with different bytes; two agent instances running against one spool
5	Whole-chain verification fails	backend, at close	No	Any gap, break or bad signature accumulated across the session

Only row 1 means the recording itself is at risk, and row 1 is detected on the consulting-room PC and never reaches the server. Rows 2 to 5 describe a consultation whose audio is complete and correct, held back by a defect in the paperwork around it. That is the case SRS-REC-07 exists to serve.

ID	Requirement	Pri	Ver	Owner
SRS- REC-01	On a server-side rejection the agent shall re-run its existing local verification (<code>read_segment</code>) and, if it passes, treat the failure as recoverable rather than terminal.	M	T	AIMS
SRS- REC-02	If the local hash matches, the segment shall be re-uploaded to a fresh object key and re-committed. The suspect object shall not be overwritten; it is evidence of the failure.	M	T	AIMS
SRS- REC-03	Retries shall be bounded — at most three, with growing delay. A rejected segment shall not be retried on every drain tick.	M	T	AIMS
SRS- REC-04	If the local hash does not match, the agent shall stop retrying verification, mark the segment unverifiable, and upload it under <code>SRS-REC-07</code> regardless. It shall not retain it.	M	T	AIMS
SRS- REC-05	The backend shall quarantine the segment , and shall quarantine the session only once retries are exhausted.	M	T	AIMS
SRS- REC-06	Quarantine shall be resolved by an automatic server-side process. Clearing it shall not require an administrator, and shall never require an action on a clinical PC.	M	T	AIMS
SRS- REC-07	A segment that cannot be verified shall still be uploaded, to a separate <code>quarantine/</code> prefix, outside the chain and explicitly marked unverifiable. It shall not enter the evidence archive, and it shall not remain on the workstation.	M	T	AIMS
SRS- REC-08	The agent shall report, in every heartbeat, the count and total bytes of segments held locally in a stuck state, so a PC hoarding unrecoverable audio is visible centrally rather than only in its own log file.	M	T	AIMS
SRS- REC-09	An operator view shall list sessions stuck in quarantine across the fleet, with clinic, room, age and size.	M	D	AIMS

On SRS-REC-07 — the distinction that matters. *Evidence* and *data* are not the same goal. The hash chain protects evidence; a consultation whose chain broke still contains clinical speech that is worth keeping, and is required downstream: a prescription cannot be generated from a recording that never arrived. Separating the two goals lets both hold. The verified archive stays strictly verified; the unverifiable recording arrives anyway, with its defect recorded against it.

7.5a.2 Nothing remains on the workstation

The consulting-room PC is a transient buffer and nothing else. It is not a place where material waits for a decision, because there is nobody there to make one — the clinician is seeing patients, and every design decision in this specification has been aimed at removing them from the loop. A file awaiting adjudication on a laptop in Ershadnagar is a file nobody will ever adjudicate.

SRS-REC-10 [M, T, AIMS] Every sealed segment shall reach the server, without exception and without human involvement. Verification determines *where* it lands — the evidence archive or the quarantine area — never *whether* it is sent.

SRS-REC-11 [M, T, AIMS] A drainage scheduler shall run independently of the live upload path, sweeping the spool on a fixed interval for anything not yet delivered — pending, failed, quarantined or orphaned by a crash — and shall deliver it. It shall not skip a segment because an earlier one failed.

SRS-REC-12 [M, T, AIMS] Local material shall be deleted once the server acknowledges receipt, whether that receipt is a purge receipt for verified audio or an acknowledgement of quarantine intake. Deletion follows custody transfer, not verification outcome.

SRS-REC-13 [M, I, AIMS] No routine or exceptional path shall require a person to open, copy, inspect or delete a file on a clinical PC. There is no supported manual intervention on a workstation.

SRS-REC-14 [M, T, AIMS] Adjudication of quarantined material shall be a server-side process that runs to a conclusion on its own — re-verify against the chain, admit to the archive if it reconciles, or file permanently as unverifiable — with a human notified of the outcome rather than blocking it.

This supersedes the earlier design. SRS-SPL-09 and the first version of this section held quarantined audio on the workstation indefinitely, pending a human decision. That was wrong on two counts. It leaves clinical audio on a laptop that AIMS LAB does not physically control, which is the opposite of the custody argument the rest of this document makes. And it withholds from the downstream pipeline exactly the recordings most likely to need attention, which becomes untenable once prescriptions are generated from that audio.

7.6 Session lifecycle and handover — SES

ID	Requirement	Pri	Ver	Owner
SRS-SES-01	Sessions shall be identified by a ULID generated by the backend.	M	I	AIMS
SRS-SES-02	A new trigger, when the gate permits, shall close the current session and open the next with no interruption in capture.	M	D	AIMS
SRS-SES-03	The new session's start time shall become the previous session's end time, so consultations tile without gap or overlap.	M	T	AIMS
SRS-SES-04	Closure and the new opening shall run on parallel threads so handover costs no recording time.	M	D	AIMS
SRS-SES-05	The archived file shall be named <code>PID_DID_HOSID_STARTTIME_ENDTIME_DATE</code> .	M	I	AIMS
SRS-SES-06	Every close shall record a reason, distinguishing normal from abnormal closure.	M	T	AIMS
SRS-SES-07	<code>superseded_by_new_patient</code> shall be a normal closure and shall not require a doctor-supplied reason.	M	T	AIMS
SRS-SES-08	Abnormal closures shall raise an integrity alert keyed on the close reason.	M	T	AIMS
SRS-SES-09	A crash mid-session shall leave a recoverable session; on restart the agent shall close it with an explicit abnormal reason rather than discarding it.	M	T	AIMS

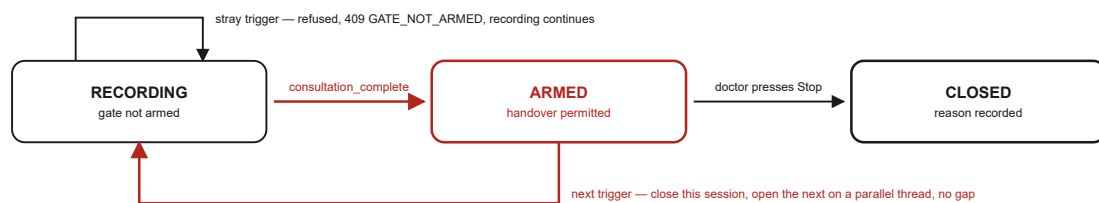
On SRS-SES-07 . Requiring a reason from the doctor at every automatic handover would demand thirty forms a day for the most ordinary event in the clinic. The reason requirement in §7.8 applies to *deliberate* stops, not to supersession.

7.7 The consultation gate — GAT

The problem. A doctor mid-consultation clicks another patient in the waiting list — to check something, or by mistake. Under the plain handover rule that trigger ends the current recording, and a consultation is cut in half with no warning and no way to know it happened.

The solution. A new trigger does not end the current session until the current session's prescription has been built.

A new patient cannot end the current recording until that consultation's prescription has been built.



Residual risk: a patient who leaves before a prescription is built never arms the gate, and the session stays open until the doctor stops it. That is why the overlay's Stop button is always visible.

Figure 7. The consultation gate. A session begins un-armed; a stray trigger is refused and the recording continues. Only the prescription-built flag arms the gate, after which the next trigger hands over with no gap in capture.

ID	Requirement	Pri	Ver	Owner
SRS-GAT-01	A session shall begin un-armed.	M	T	AIMS
SRS-GAT-02	Only <code>consultation_complete</code> for the open session's patient shall arm the gate.	M	T	AIMS
SRS-GAT-03	A trigger arriving while un-armed shall be refused <code>409 GATE_NOT_ARMED</code> ; the current recording shall continue undisturbed.	M	T	AIMS
SRS-GAT-04	A trigger arriving while armed shall perform the handover of SRS-SES-02 .	M	T	AIMS
SRS-GAT-05	The overlay Stop button shall close a session regardless of gate state, subject to §7.8.	M	T	AIMS
SRS-GAT-06	Every refused trigger shall be logged with patient, doctor and time, and surfaced in operational reporting.	M	T	AIMS
SRS-GAT-07	Gate state shall be reported in <code>status</code> so CMED can display it.	S	T	AIMS

Known residual risk. If a patient leaves before a prescription is built — they were sent for an investigation, they walked out, the doctor referred them — the gate never arms, and the session stays open until the doctor stops it or the day ends. This is precisely why the overlay's Stop button exists and

why it is always visible. **OD-04** asks CMED and the clinical team what share of consultations end without a prescription, because that number decides whether an inactivity timeout is also needed.

7.8 The recording-control overlay — **UIX**

Today a doctor who wants to pause must find the system tray, locate AIMScribe, right-click, and choose from a menu — during a consultation, with a patient watching. That is not usable, and an unusable control is an unused control.

The replacement. When recording starts, a small always-on-top window appears at the top right of the screen, in front of the browser. It is not draggable and never needs to be: it is small, fixed, and positioned to overlap nothing clinically important.

Control	Shape	Colour	Action
Stop	Circle	Red	Ends the session — after a reason
Pause	Rectangle	Blue	Suspends capture — after a reason

ID	Requirement	Pri	Ver	Owner
SRS- UIX-01	The overlay shall appear automatically when a session starts and disappear when it closes.	M	D	AIMS
SRS- UIX-02	It shall be always-on-top, fixed at the top right, and shall not require dragging.	M	D	AIMS
SRS- UIX-03	It shall present exactly two controls: a red circular Stop and a blue rectangular Pause.	M	I	AIMS
SRS- UIX-04	Pressing Pause shall present a list of reasons plus a free-text comment box.	M	D	AIMS
SRS- UIX-05	The pause shall take effect only after a reason is chosen or typed and confirmed.	M	T	AIMS
SRS- UIX-06	Pressing Stop shall present the same form, and the session shall close only after a reason is chosen or typed and confirmed.	M	T	AIMS
SRS- UIX-07	The reason form shall not be dismissable without either supplying a reason or cancelling the action entirely.	M	T	AIMS
SRS- UIX-08	On Stop, the microphone shall cut immediately on the button press; the reason shall govern how the session is <i>filed</i> , not whether the doctor stops being recorded.	M	T	AIMS
SRS- UIX-09	Reasons shall be recorded in the chain as signed <code>pause</code> / <code>resume</code> / <code>close</code> entries.	M	T	AIMS
SRS- UIX-10	A pause exceeding a configured duration shall be flagged for review.	S	T	AIMS
SRS- UIX-11	The overlay shall be operable by keyboard and legible at 125 % and 150 % display scaling.	S	D	AIMS
SRS- UIX-12	The overlay shall require nothing of CMED.	M	I	AIMS

On SRS-UIX-08 — the subtlety that took a round of discussion to settle. The requirement is that a doctor cannot stop a recording without giving a reason, so that a vague or absent justification cannot quietly damage data collection. But if the doctor has pressed Stop because the patient just objected to being recorded, the microphone must stop *now*, not after a form is filled in. The resolution: the press cuts the microphone; the form governs the session's filing. The doctor cannot escape the form — the session stays open and unfilled until it is completed — but the patient's objection is honoured instantly.

7.9 Backend — BKD

ID	Requirement	Pri	Ver	Owner
SRS-BKD-01	Consent shall be enforced independently in three places: the mint, the agent, and a database CHECK constraint.	M	T	AIMS
SRS-BKD-02	An <code>audit_log</code> shall record every security-relevant event, append-only, enforced by a database trigger rather than convention.	M	T	AIMS
SRS-BKD-03	Every write path shall be idempotent under retry; a repeated commit shall not create a duplicate segment row.	M	T	AIMS
SRS-BKD-04	Identifiers shall be validated against <code>^[A-Za-z0-9_-]{1,64}\$</code> at every boundary.	M	T	AIMS
SRS-BKD-05	Error responses shall not disclose which of several failure causes occurred where that would help an attacker.	M	I	AIMS
SRS-BKD-06	The backend shall expose a health endpoint reporting database and object-storage reachability.	M	D	AIMS

On SRS-BKD-04 . Identifiers become folder names on the archive volume. Keep them boring — this single regex is what stands between a stray `../` and the archive tree.

7.10 Archive worker — ARC

ID	Requirement	Pri	Ver	Owner
SRS-ARC-01	The worker shall archive only sessions whose chain has been verified.	M	T	AIMS
SRS-ARC-02	It shall concatenate segments in sequence order into one WAV per consultation.	M	T	AIMS
SRS-ARC-03	It shall verify the concatenated result before reporting completion.	M	T	AIMS
SRS-ARC-04	It shall file under <code>hospital_id</code> as the top-level folder.	M	I	AIMS
SRS-ARC-05	It shall issue purge receipts only after the archive copy is verified.	M	T	AIMS
SRS-ARC-06	It shall poll on a bounded interval with a bounded batch, and shall degrade gracefully when the backend is unreachable.	M	T	AIMS
SRS-ARC-07	The archive volume shall be backed up, and the backup shall be restore-tested.	M	D	AIMS

8. Data requirements

8.1 Data crossing the CMED boundary

CMED → AIMScribe

Datum	Classification	Retained?
patient_id	Pseudonymous identifier	Yes — filename and database
doctor_id	Identifier	Yes
clinic_id	Identifier	Yes, after mapping
start_time , date	Timestamp	Yes
consent_obtained , consent_method	Consent record	Yes
patient_name	Personal data	Yes — from the clinical record (§8.6)
Demographics	Personal data	Yes — at trigger
Prescription contents	Clinical	Yes — at conclusion
Diagnoses, notes, history	Clinical	Yes — at conclusion
Previous prescription	Clinical	Yes — at trigger, for a returning patient

AIMScribe → CMED

Datum	Notes
status , code , message	Outcome of a command
session_id	Opaque ULID
Recording state, gate state	For the indicator
Warnings	Disk, microphone, connectivity
Audio	Never. Not in any form, at any time.
Transcripts or AI output	Never through this interface

SRS-DAT-01 [M, I, AIMS] Audio shall never be transmitted to CMED, in any form, through any interface described in this document.

SRS-DAT-02 [M, T, AIMS] patient_name shall never appear in a chain entry, a filename, an archive path, an object key or a log line. It is stored only as a column of the clinical record (§8.7), where access is separately controlled.

SRS-DAT-03 [M, I, Joint] Clinical content crosses the boundary in one direction only: CMED to AIMS LAB. Nothing clinical is ever returned.

This is a reversal, and it must be recorded as one. Versions 1.0 to 1.5 of this specification stated that prescriptions, diagnoses, notes and patient names would never cross the boundary, and the design derived several properties from that promise — most visibly the rule that identifiers are "boring"

because they become directory names carrying nothing sensitive. Ingesting the clinical record (§8.6) changes the classification of the whole archive volume, not merely of the new files. Three consequences follow and are specified rather than assumed:

Was true while only audio was held	Is true once the clinical record is held
An archive directory name disclosed nothing	The directory now sits beside a file naming the patient
Consent covered audio recording	Consent must cover extraction of the clinical record
Loss of the volume exposed unlabelled audio	Loss of the volume exposes an identified clinical record

SRS-DAT-15 [M, I, Joint] The consent text presented to patients shall be revised before any clinical record is ingested, and the revision shall be recorded with a date. Consent obtained for audio recording alone does not extend to extraction of the record.

SRS-DAT-16 [M, I, AIMS] The archive volume and the clinical database shall be encrypted at rest, and access to the clinical database shall be granted separately from access to the audio catalogue. A role able to search recordings shall not thereby be able to read prescriptions.

8.2 Identifier rules

SRS-DAT-04 [M, T, Joint] All identifiers shall match `^[A-Za-z0-9_-]{1,64}$`.

SRS-DAT-05 [M, I, CMED] Identifiers shall be stable over time. If `patient_id` for a given patient changes, that patient's recordings become two unrelated sets and cannot be reconciled afterwards.

SRS-DAT-06 [M, I, CMED] Identifiers shall not embed personal data. A patient identifier containing a name or a national ID number places that data in filenames, logs and metrics.

8.3 Retention

Data	Location	Retention
Spooled encrypted segments	Consulting-room PC	Until purge receipt + 24 h
Segment objects	Object storage	Until archived and verified
Archived WAV	AIMS LAB archive volume	OD-06 — undecided
Session metadata	Database	Indefinite
Audit log	Database	Indefinite, append-only
Agent logs	PC	30 days

SRS-DAT-07 [M, A, AIMS] A raw-audio retention period shall be decided before hardware is purchased. At 11 TB/year, retention is the single largest cost driver in the system. **OD-06**.

8.4 Data protection

SRS-DAT-08 [M, I, AIMS] Audio shall be encrypted at rest at every stage: AES-256-GCM in the spool, server-side encryption in object storage, and encrypted volumes in the archive.

SRS-DAT-09 [M, I, AIMS] Audio shall be encrypted in transit at every stage: TLS 1.2+ throughout.

SRS-DAT-10 [M, I, AIMS] No secret shall appear in either public source repository. `.env` shall be gitignored, and enrolment token sheets treated as credentials.

8.5 One original, one derivative

Improving audio for recognition and preserving it as evidence are incompatible demands on a single file. They are not reconciled by compromise; they are reconciled by keeping two renditions.

	Archive — the original	ASR rendition — derived
Processing	none, ever	whatever aids recognition
Hashed into the chain	yes	no
Retained	per retention policy	disposable, regenerable
Format	WAV PCM, capture rate, mono	16 kHz mono

SRS-DAT-11 [M, I, AIMS] The archived recording shall be bit-identical to the captured audio. No filtering, levelling, noise reduction or re-encoding shall ever be applied to it.

SRS-DAT-12 [M, T, AIMS] Any rendition prepared for speech recognition shall be derived from the archived original, stored separately, and shall never replace it or be hashed as evidence.

SRS-DAT-13 [M, I, AIMS] The processing chain and its parameters shall be versioned and recorded, so a transcript can be traced to exactly how its audio was prepared and the rendition can be regenerated.

SRS-DAT-14 [S, A, AIMS] The recognition chain shall avoid aggressive spectral noise reduction and fast-acting compression, both of which introduce artefacts that increase hallucination in transformer-based recognisers. Measured room noise concentrates at 100–300 Hz, so a high-pass at 80 Hz and long-window level normalisation carry most of the benefit.

8.6 Clinical record ingestion — CRI

The record is required at two distinct moments, and the interface already has a signal at each of them. Nothing new is added to the transport.

Moment	Existing signal	What CMED sends	Why then
Consultation begins	<code>start</code> (the trigger)	Demographics captured at reception; for a returning patient, the most recent previous prescription in full	The clinician needs the history at the start; a record delivered afterwards cannot inform the encounter it describes
Consultation concludes	<code>consultation_complete</code>	Prescription contents, diagnoses, notes, investigations	These do not exist until the prescription is built — which is the event that already arms the gate

- SRS-CRI-01** [M, T, CMED] The trigger shall carry a `demographics` object holding the data captured before the patient entered the consulting room.
- SRS-CRI-02** [M, T, CMED] For a patient with a prior encounter, the trigger shall additionally carry `previous_prescription` : the complete contents of the most recent one, with its date.
- SRS-CRI-03** [M, T, CMED] `consultation_complete` shall carry the prescription contents, diagnoses, clinical notes and investigation orders for the encounter just concluded.
- SRS-CRI-04** [M, T, AIMS] Ingestion shall never block acquisition. A record that is absent, late, malformed or rejected shall be logged and reconciled afterwards; the recording proceeds regardless.
- SRS-CRI-05** [M, T, AIMS] Each record shall be written to the archive beside its audio, in the same directory, under the same stem with a `.json` extension — `PID_DID_HOSID_START_END_DATE.json`.
- SRS-CRI-06** [M, T, AIMS] The record shall also be ingested into the clinical database (§8.7) field by field. The file is the artefact; the database is the index. Neither is derived from the other at query time.
- SRS-CRI-07** [M, T, AIMS] A record shall be validated against a published schema on arrival. A record failing validation shall be stored unmodified in a quarantine area and reported, never silently discarded and never partially ingested.
- SRS-CRI-08** [M, I, AIMS] `patient_name` shall be taken from the record and stored only in the clinical database. It shall not enter the filename, the directory path, the chain or the audio catalogue.
- SRS-CRI-09** [M, T, AIMS] Ingestion shall be idempotent. A record delivered twice shall produce one row, not two.
- SRS-CRI-10** [M, T, AIMS] The record and its audio shall be reconciled nightly. A recording with no record, and a record with no recording, shall both be reported — the two are different faults with different causes.
- SRS-CRI-11** [S, I, Joint] The set of fields required at each moment shall be agreed in writing and versioned. A field added later shall not invalidate records already stored under an earlier version.
- SRS-CRI-12** [M, I, AIMS] Records shall be encrypted at rest and shall not be transmitted to any downstream service that does not require them.

Reconciliation is the requirement to take seriously. Audio arrives over a store-and-forward path that tolerates three weeks of disconnection; the record arrives over a live browser channel that tolerates none. The two will therefore diverge routinely, and a nightly reconciliation is the only way that divergence becomes visible rather than accumulating.

8.7 Database architecture — DBA

One database. Two schemas. Separation of access is required; separation of databases is not the way to get it, and costs the same things it cost in §8.7.2 — two connections, two migration paths, and a dashboard that cannot join across the line it needs to report on.

Schema	Holds	Who may read it
audio	Sessions, segments, chain entries, devices, alerts	Operations, research engineering
clinical	Demographics, prescriptions, diagnoses, notes, patient names	A separate, narrower role

SRS-DBA-20 [M, I, AIMS] Both domains shall live in one database, in two schemas, with privileges granted per schema. A role holding `USAGE` on `audio` and none on `clinical` can search every recording and run the dashboard without being able to read a patient name.

SRS-DBA-21 [M, T, AIMS] Cross-schema joins shall be available to roles holding both, so that reconciliation and reporting are single queries.

This gives the isolation the two-database design was reaching for, enforced by the database rather than by which connection string an application happened to open, and it keeps one migration path.

8.7.1 The audio catalogue

The reporting hierarchy is clinic → clinician → date → patient, and the schema is indexed for exactly that traversal, because it is what both the dashboard and manual retrieval ask for.

SRS-DBA-01 [M, I, AIMS] The catalogue shall model `hospitals`, `doctors`, `devices`, `sessions`, `segments`, `chain_entries` and `integrity_alerts`, with foreign keys enforced rather than implied.

SRS-DBA-02 [M, T, AIMS] The hierarchy shall be served by a composite index on `(hospital_id, doctor_id, session_date)`, extended to include `patient_id`.

SRS-DBA-03 [M, T, AIMS] Every session row shall carry its archive filename and relative path, so a recording is locatable by query alone.

Why sixty recordings feel slow, which is not what it looks like. The indexes implied above already exist — `idx_sessions_archive_lookup` on `(hospital_id, doctor_id, session_date)`, `idx_sessions_patient_date`, and a unique index on `archive_relpath`. Sixty rows would be fast without any of them. The cost is therefore not in the query, and three other causes are:

Cause	Why it presents as a slow search	Fix
Two disconnected catalogues	Session metadata lives in Postgres; the archived files are indexed in a separate SQLite catalogue on the archive server. Neither alone answers "where is this recording", so the search becomes a manual walk of <code>hospital/doctor/date/</code> directories	SRS-DBA-22
No query interface	There is no dashboard over either catalogue, so "search" means opening a folder or writing SQL by hand	\$8.8
Database autosuspend	A database that scales to zero spends several seconds waking before answering anything. Against sixty rows, that wait is the entire perceived query time	SRS-DBA-23

SRS-DBA-22 [M, T, AIMS] There shall be one authoritative catalogue. The archive worker shall write the archived filename, relative path, byte count and digest back into it on completion, so that a single query answers where a recording is and whether it is verified. A second catalogue on the archive server may exist as a local convenience but shall not be authoritative.

SRS-DBA-23 [M, I, AIMS] Autosuspend shall be disabled on the operational database. A cold start on a clinical path is not an acceptable latency, and it is indistinguishable to a user from a slow query.

SRS-DBA-24 [S, T, AIMS] Substring search on a filename shall be supported by a trigram index (`pg_trgm`), because a B-tree cannot serve a leading wildcard. Without it, filename search degrades to a sequential scan as the corpus grows — invisible at sixty rows, disabling at sixty thousand.

SRS-DBA-25 [M, T, AIMS] The filename components — patient, doctor, hospital, start, end, date — shall each be stored as their own column and not parsed out of the filename at query time. The filename is a label for humans; the columns are what queries use.

SRS-DBA-04 [M, T, AIMS] The catalogue shall record, per session, the close reason, whether the chain verified, whether it was quarantined and with what reason, and whether it was terminated by the clinician.

8.7.2 The clinical record

On separating male and female patients into different databases — a recommendation against, with the alternative. The requirement driving it is real: the required fields differ by sex. But separate databases would break the dashboard requested in §8.8 of this same specification. "How many patients were seen at Dholpur today" would become two queries against two databases, unioned in application code, and any future report that forgets one side would under-count silently rather than fail. Migrations would have to be applied twice and would drift. A patient whose recorded sex is corrected would have to be moved between databases, losing the primary key guarantee and every foreign key pointing at it.

Differing fields are a schema question, not a database question, and Postgres answers it directly:

SRS-DBA-05 [M, I, AIMS] Patients shall be held in one table in one database. Sex-specific data shall be held in dedicated satellite tables joined on `patient_id` — for example an obstetric history table populated only for the patients it applies to.

SRS-DBA-06 [M, T, AIMS] Sex-conditional mandatory fields shall be enforced by table-level `CHECK` constraints, so that a record missing a field its sex requires is rejected by the database rather than by convention.

This gives per-sex validation, single-query reporting, one migration path, and a correction that is an `UPDATE` rather than a migration. If separate databases are nonetheless preferred, the schema below transfers unchanged and §8.8 acquires a union in every query; that trade should be made deliberately.

On current and previous prescriptions as two tables — the same reasoning applies more sharply. Two tables keyed on `patient_id` can hold exactly one previous prescription, so the third visit silently destroys the first. That is data loss by schema.

SRS-DBA-07 [M, I, AIMS] Prescriptions shall be stored in one table keyed on `(patient_id, encounter_id)`, retaining every encounter.

SRS-DBA-08 [M, T, AIMS] `current_prescription` and `previous_prescription` shall be provided as **views** over that table, resolved by encounter date. The two-table interface is preserved exactly; the history behind it is not discarded.

SRS-DBA-09 [M, T, AIMS] `previous_prescription` served at trigger time shall be the most recent encounter **strictly before** the current one, so that a same-day repeat visit cannot return the encounter in progress.

Structure. Prescriptions shall be stored field by field, not as an opaque blob:

SRS-DBA-10 [M, I, AIMS] Prescription contents shall be normalised into typed columns and a `prescription_items` child table — one row per prescribed item, with drug, dose, frequency and duration as separate columns. Free text shall remain free text; structured data shall not.

SRS-DBA-11 [S, I, AIMS] The original JSON shall be retained alongside the normalised form in a `JSONB` column, so that a field not yet modelled is not lost and can be extracted later without re-ingestion.

8.7.3 What makes it industry grade

SRS-DBA-12 [M, I, AIMS] Schema changes shall be applied only by versioned, forward-only migrations held in version control and applied by CI. No schema change shall be made through a web console.

SRS-DBA-13 [M, T, AIMS] Constraints shall be expressed in the database — foreign keys, `NOT NULL`, `CHECK`, uniqueness — and not solely in application code.

SRS-DBA-14 [M, T, AIMS] Every table carrying clinical or identifying data shall have `created_at` and `updated_at`, maintained by trigger.

SRS-DBA-15 [M, T, AIMS] Access to the clinical database shall be by role, with read, write and administrative roles distinct, and no application connecting as the owner.

SRS-DBA-16 [M, D, AIMS] Point-in-time recovery shall be enabled, and a nightly encrypted logical dump shall be written to a different provider. Point-in-time recovery protects against an operator's mistake; it does not protect against losing the account.

SRS-DBA-17 [M, D, AIMS] Restoration shall be tested on a schedule, and the test recorded. An untested backup is a belief, not a control.

SRS-DBA-18 [M, T, AIMS] Reads and writes of clinical data shall be written to an append-only audit log recording who, what and when.

SRS-DBA-19 [S, I, AIMS] Connection pooling shall be in transaction mode, and the driver configured accordingly — server-side prepared statements are unavailable in that mode and fail obscurely if assumed.

8.8 Operational dashboard — DSH

SRS-DSH-01 [M, D, AIMS] The dashboard shall report daily volume by clinic and by clinician, as counts of recordings and total recorded hours, shown both as a chart and as a table.

SRS-DSH-02 [M, D, AIMS] It shall support the hierarchy of §8.7.1 — clinic, then clinician, then date, then patient — and allow retrieval of a recording by any of them.

SRS-DSH-03 [M, D, AIMS] It shall report integrity outcomes: recordings whose chain failed verification, recordings quarantined and under which of the five triggers, and recordings terminated by the clinician before the encounter concluded.

SRS-DSH-04 [M, D, AIMS] It shall report acquisition health per room: stuck local bytes (**SRS-REC-08**), speech-level outcomes (**SRS-LVL-02**), and agents not seen within three heartbeat intervals.

SRS-DSH-05 [M, D, AIMS] It shall report reconciliation state: recordings without a clinical record, and records without a recording (**SRS-CRI-10**).

SRS-DSH-06 [M, I, AIMS] The dashboard shall read the audio catalogue only. It shall not display patient names or clinical content, so that operational monitoring does not require clinical access.

SRS-DSH-07 [S, D, AIMS] Counts shall be derivable to their underlying rows. A figure a reader cannot drill into is a figure they cannot act on.

SRS-DSH-03 is the one that changes behaviour rather than reporting it. A quarantined session or a forced termination is presently visible only to whoever reads an alert table or a log file on the machine itself; surfacing both against clinic and clinician makes a recurring problem in one room visible as a pattern rather than as a series of unrelated incidents.

9. Non-functional requirements

9.1 Performance

ID	Requirement	Pri	Ver
SRS-NFP-01	Trigger to microphone open: ≤ 500 ms at the 95th percentile.	M	A
SRS-NFP-02	Trigger to 200 RECORDING_STARTED : ≤ 2 s at p95 on a working link.	M	A
SRS-NFP-03	Handover between patients: no gap in capture, measured on a continuous test tone.	M	T
SRS-NFP-04	Agent CPU: < 5 % of one core, average, while recording.	S	A
SRS-NFP-05	Agent memory: < 250 MB resident.	S	A
SRS-NFP-06	Backend p95 latency: < 300 ms for control routes, excluding segment commit.	S	A
SRS-NFP-07	Segment commit p95: < 3 s including read-back and re-hash.	S	A

9.2 Capacity

ID	Requirement	Pri	Ver
SRS-NFC-01	7 clinics, 14 consulting rooms, 30 doctors across two shifts, ~ 560 sessions/day, ~ 112 recording-hours/day. Peak concurrency is 14, bounded by rooms rather than doctors.	M	A
SRS-NFC-02	Sustain ≈ 36 GB/day ingest — 0.9 TB/month, 11 TB/year.	M	A
SRS-NFC-03	Absorb a $3\times$ burst for one hour without loss, by spooling.	M	A
SRS-NFC-04	Support $2\times$ growth without architectural change.	S	A

9.3 Availability and reliability

ID	Requirement	Pri	Ver
SRS-NFR-01	The agent shall record with the backend entirely unreachable, for up to three weeks of one room's work.	M	D
SRS-NFR-02	Backend availability target: 99.5 % monthly, excluding planned maintenance.	S	A
SRS-NFR-03	A backend deployment shall not interrupt any recording in progress.	M	D
SRS-NFR-04	No single component failure shall cause silent loss of audio; every loss path shall raise an alert.	M	T
SRS-NFR-05	Recovery point objective for archived audio: zero.	M	A

9.4 Security

ID	Requirement	Pri	Ver
SRS-NFS-01	Defence shall be layered: origin, Host, loopback, grant, device token, chain, read-back verification, purge receipt. No single layer's failure shall be sufficient.	M	I
SRS-NFS-02	Bearer-token comparison shall be constant-time.	M	I
SRS-NFS-03	AIMScribe_Agent.exe and AIMScribeSetup.exe shall be Authenticode-signed.	M	I
SRS-NFS-04	Every credential shall be rotatable without reinstalling agents, except the device key, which requires re-enrolment by design.	S	D
SRS-NFS-05	Security-relevant failures shall be logged with enough context to diagnose, and without disclosing secrets.	M	I
SRS-NFS-06	The system shall assume the browser page is hostile and derive no trust from its claims.	M	I

9.5 Observability

ID	Requirement	Pri	Ver
SRS-NFO-01	Every agent shall heartbeat every 30 s with version, spool depth and pending count.	M	T
SRS-NFO-02	An agent silent for more than 3 heartbeat intervals during clinic hours shall alert.	M	T
SRS-NFO-03	Alerts shall exist for: chain verification failure, hash mismatch on arrival, spool above 80 %, abnormal close, silent session, repeated gate refusal, archive backlog growth.	M	T
SRS-NFO-04	An operator dashboard shall show fleet state without database access.	S	D
SRS-NFO-05	Logs shall be structured and free of patient identifiers beyond the pseudonymous reference.	S	I

9.6 Maintainability and portability

ID	Requirement	Pri	Ver
SRS-NFM-01	Wire protocol version shall be explicit; a mismatch shall be refused, never guessed at.	M	T
SRS-NFM-02	Golden wire vectors with pinned SHA-256 shall exist as a cross-repository compatibility tripwire.	M	T
SRS-NFM-03	The backend shall be deployable to a managed platform or to the AIMS LAB server from the same container image.	M	D
SRS-NFM-04	No component shall depend on a provider-proprietary API without an abstraction permitting substitution.	S	I
SRS-NFM-05	Unknown fields in a received message shall be ignored, not rejected, so either side can add fields without breaking the other.	M	T

9.7 Usability

ID	Requirement	Pri	Ver
SRS-NFU-01	A doctor shall need no training beyond "the red circle stops it, the blue rectangle pauses it".	M	D
SRS-NFU-02	Installation shall be one signed installer, three fields, two pre-filled.	M	D
SRS-NFU-03	The tray icon shall convey recording state at a glance.	S	D
SRS-NFU-04	Every message shown to a doctor shall be plain language with no technical vocabulary.	M	I

10. Capacity and sizing

10.1 The load model

Parameter	Value	Source
Clinics	7	6 Aalo clinics + Amader Susastho
Consulting rooms	14	Deployment plan
Peak concurrent recordings	14	A room records one consultation at a time
Enrolled laptops	16	One per room, plus two spares
Doctors	30	Two shifts share the same rooms
Room hours per day	8	Morning and afternoon shift
Recording-hours per day	112	14 rooms × 8 h
Mean consultation	12 min	Estimate — OD-05
Sessions per day	~560	Derived
Audio bitrate	88,200 B/s	SRS-CAP-01
Per recorded hour	318 MB	Derived

10.2 Derived load — and the conclusion that reframes the question

Quantity	Value
Segments per day	~9,000
API requests per day	~28,000
Mean request rate	~1.2 req/s
Peak request rate (3×)	~4 req/s
Daily ingest	~36 GB
Monthly	~0.9 TB
Annual	~11 TB
API tier throughput incl. read-back	~2.5 MB/s ≈ 20 Mbit/s

This is not a compute-bound system. It is a storage-and-bandwidth-bound one.

A little over one request per second is trivial — a single modest instance handles it with room to spare. 36 GB per day is not trivial, and it arrives every working day whether anyone is watching.

Size this system by rooms, not by doctors. A consulting room records one consultation at a time, so 30 doctors sharing 14 rooms across two shifts produce 14 concurrent recordings, not 30. Every figure above follows from the room count.

Do not size the API tier from the request rate alone. Because `SRS-UPL-02` re-reads and re-hashes every segment server-side, the API tier carries the full audio volume in both directions. It needs about 20 Mbit/s of sustained throughput, not the near-zero that 1.2 req/s would suggest. It also holds one segment in memory per in-flight commit, and hashing runs on a bounded thread pool — so instance memory and pool size are real parameters, not defaults to leave alone.

10.3 Cloud deployment

Component	Specification	Rationale
API	2 × (1 vCPU, 2 GB RAM)	One handles the load. Two exist for zero-downtime deploys and for one failing.
Database	Managed Postgres, 1–4 vCPU autoscaling, 100 GB, autosuspend off	Metadata is small; suspend adds cold-start latency to a clinical path
Object storage	S3-compatible, no egress fees	Egress fees on 11 TB/year dominate every other cost
Cache / queue	Redis, 256 MB	Rate limits, job coordination
Estimated	USD 120–200/month	Excludes archive storage and the out-of-scope AI pipeline

SHA-NI hardware hashing is present on any current x86 instance and makes re-hashing effectively free relative to network cost — worth confirming rather than assuming when choosing an instance family.

10.4 On-premises deployment at AIMS LAB

Component	Specification	Rationale
CPU	16 cores	8 would serve; 16 leaves room for the AI pipeline on the same box
RAM	64 GB ECC	ECC is not optional for a system whose purpose is detecting corruption
Database disk	500 GB NVMe with power-loss protection	See below
Bulk storage	8 × 12 TB, RAIDZ2 → ~60 TB usable	~5 years at 11 TB/year
Network	50–100 Mbit/s symmetric	See below
Power	UPS with clean shutdown	See below
OS	Linux, containerised deployment	Same image as cloud (<code>SRS-NFM-03</code>)

Three numbers decide whether this build succeeds, and two of them are not capacity.

Power-loss protection is mandatory. For the database disk, the number that matters is fsync latency, not capacity: target p99 under 1 ms. Consumer SSDs without PLP *lie about flush completion* — they acknowledge a write that is still in a volatile cache. That is precisely the failure this entire architecture exists to detect. A system that builds a signed hash chain on top of a disk that lies is not an evidence system.

The uplink must be symmetric. This is an upload-dominated workload — 42 Mbit/s sustained inbound to the server. Most business connections in Bangladesh are asymmetric, advertising a download figure that is irrelevant here.

The UPS is a durability requirement, not a convenience. Without clean shutdown, every claim about write durability in §7.3 is unproven at exactly the moment it matters.

10.5 Per-clinic network

Rooms at site	Sustained upload needed
2 (the average across 7 sites)	~1.4 Mbit/s
3	~2.1 Mbit/s
4	~2.8 Mbit/s

Modest — but sustained for six hours, and the spool covers the gaps rather than the link being sized for the peak.

10.6 Growth

Scale	Rooms	Daily	Annual	What changes
Today	14	36 GB	11 TB	—
2×	28	71 GB	22 TB	Storage only
5×	70	178 GB	55 TB	Storage, plus a second API instance genuinely needed
10×	140	356 GB	111 TB	Storage tiering; regional deployment

At every scale the first constraint is archive retention, not compute. This is **SRS-DAT-07** and **OD-o6** restated as a growth curve.

11. Failure modes and degraded operation

#	Failure	Detection	Behaviour	Doctor sees	Clinical impact
1	Agent not installed	Connection refused	CMED logs and continues	Nothing	None
2	Agent stopped	Connection refused	CMED retries with backoff	Nothing	None
3	Backend unreachable	Request timeout	Record and spool locally; up to 3 weeks	Tray warning only	None
4	Clinic internet down	Same as 3	Same as 3	Tray warning only	None
5	Spool above 80 %	Local check	Alarm raised; recording continues	Tray warning	None yet
6	Spool full / < 20 GB free	Local check	Refuse to start new sessions	Clear message	Recording lost, consultation unaffected
7	Microphone lost mid-session	Capture error	Warn, attempt recovery, preserve what exists	Overlay warning	Partial recording
8	Grant refused	Backend reply	No recording; captured audio discarded	Brief message	Recording lost
9	Clock skew > 65 s	Grant expiry fails	Recording refused	"Authorisation failed"	Recording lost until clock fixed
10	Clinic mismatch	Backend check	Refuse and alert (SRS-GRT - 10)	Clear message	Recording lost — deliberately
11	Hash mismatch on arrival	Read-back verify	Quarantine session, alert, never auto-delete	Nothing	Investigated, not lost
12	Chain verification fails at close	Backend verify	Quarantine, alert	Nothing	Investigated
13	Power loss mid-segment	Journal on restart	Partial segment not accepted; ≤ 99 s of audio lost	Nothing	Bounded loss
14	Flag never arrives	Gate stays un-armed	Session stays open; Stop button available	Overlay visible	Long session, SRS-GAT-06 reporting
15	Two triggers in rapid succession	Gate state	Second refused while un-armed; handover once armed	Nothing	None
16	Archive worker down	Backlog growth	Segments retained; nothing purged without a receipt	Nothing	None — deferred

Read column six. In sixteen failure modes, the clinical impact is *none* in nine and *a lost recording* in the rest. In none of them does a doctor fail to see a patient. That is SRS-IF1-17 holding all the way down, and it is the property to point at when someone asks what happens when this breaks.

Part IV — Closing out

12. Verification and acceptance

12.1 Acceptance tests

Each test is pass/fail on a running system, with the requirements it verifies.

ID	Test	Verifies	Pass criterion
AT-01	Doctor opens a patient in CMED	IF1-01, CAP-03, GRT-07	Microphone opens within 500 ms; 200 RECORDING_STARTED within 2 s
AT-02	Trigger with doctor_id omitted	INV-03	Refused 400 MISSING_FIELD ; nothing recorded
AT-03	Trigger with consent_obtained: false	BKD-01	Refused 400 CONSENT_REQUIRED
AT-04	Trigger naming a clinic other than the device's	GRT-10, ENR-20	Refused 401 CLINIC_MISMATCH ; alert raised
AT-05	Connect from a non-allowlisted origin	IF1-06	Socket closed 4403 before the handshake completes
AT-06	Connect with Host: evil.example resolving to 127.0.0.1	IF1-06	Refused — DNS-rebinding defence
AT-07	Replay a captured grant inside 60 s	GRT-05	Refused
AT-08	Present a grant 90 s old	GRT-04	Refused on expiry
AT-09	Trigger for patient B while recording patient A, gate un-armed	GAT-03	409 GATE_NOT_ARMED ; patient A's recording continues uninterrupted
AT-10	Send consultation_complete , then trigger patient B	GAT-04, SES-02, SES-03	Session A closes, B opens, continuous test tone shows no gap
AT-11	consultation_complete naming the wrong patient	IF1-13	409 PATIENT_MISMATCH ; gate not armed
AT-12	consultation_complete sent twice	IF1-14	Second returns 200 GATE_ALREADY_ARMED
AT-13	Press overlay Stop, dismiss the reason form	UIX-06, UIX-07	Form cannot be dismissed; session does not close
AT-14	Press overlay Stop, supply a reason	UIX-06, UIX-08	Microphone cuts on press; session closes on confirm; reason in the chain
AT-15	Press overlay Pause without a reason	UIX-05	Pause does not take effect
AT-16	Disconnect the clinic's internet for 4 h of recording	NFR-01, UPL-06	All audio spooled; uploads complete on restoration; no loss
AT-17	Pull power mid-segment	SPL-05, SPL-04	Partial segment rejected on restart; journal consistent; loss ≤ 99 s
AT-18	Corrupt a segment in object storage before commit	UPL-02, UPL-03	Backend detects on read-back; session quarantined
AT-19	Remove one chain entry before close	CHN-05	Verification fails; session quarantined
AT-20	Fill the spool to 80 %, then to the refusal threshold	SPL-07	Alarm at 80 %; new sessions refused below 20 GB free

ID	Test	Verifies	Pass criterion
AT-21	Redeem an enrolment token twice from the same never-seen device	ENR-08	Second redemption succeeds with re-issued credentials
AT-22	Redeem a token twice after the device has heartbeated	ENR-08	Refused; replay logged
AT-23	Redeem an expired token	ENR-03, ENR-04	Refused with the same message as unknown and used
AT-24	Enrol with the backend unreachable	ENR-07	Token remains on disk; retry succeeds later
AT-25	Wipe the key folder and restart	ENR-11	Agent refuses to record and demands re-enrolment
AT-26	Revoke a device, then attempt to upload	ENR-15, ENR-16	Refused at every route
AT-27	Stop the agent, then trigger from CMED	IF1-17	CMED logs and continues; no dialog; doctor unaffected
AT-28	Set the PC clock 10 min fast	ASM-07, GRT-04	Recording refused; log names clock skew as the cause
AT-29	Run 14 simulated rooms recording concurrently for one clinic-day	NFC-01, NFP-06	No dropped segments; p95 latencies within \$9.1
AT-34	Record 60 s of occupied room and count frames at digital zero	CAP-11	Below 0.5%; above that the room is not commissioned
AT-35	Compare capture through every available host API on the same room	CAP-07, CAP-08	Native-rate low-latency path is chosen and is the least gated
AT-36	Speak loudly from the doctor chair, quietly from the patient chair	LVL-01, LVL-03	Quiet-talker prompt fires; loud-talker prompt does not
AT-37	Mute the microphone at the hardware switch mid-session	LVL-04, CAP-06	Reports a microphone problem, not a patient problem
AT-38	Record a clinic day in a commissioned room and bucket segment lengths	SEG-07	At least 80% within 30–60 s
AT-39	Feed a recording whose speech sits below the fixed silence constant	SEG-06	Segments still land in the target window; no 15 s collapse
AT-40	Re-enrol a device with segments still in its spool	ENR-21	Refused until the spool is drained
AT-41	Compare the archived WAV byte-for-byte against the captured stream	DAT-11	Identical; SHA-256 matches the chain
AT-42	Corrupt a segment in transit, leaving the local file intact	REC-01, REC-02	Local re-hash matches; re-uploaded to a fresh key; session completes
AT-43	Corrupt the local segment file on disk, then let it upload	REC-01, REC-04	Retries stop; alert names local media corruption; file kept
AT-44	Exhaust the retry budget on one segment	REC-03, REC-05	Exactly three attempts; session quarantines only after the third

ID	Test	Verifies	Pass criterion
AT-45	Clear a quarantine after a verified re-upload	REC-06	Session archives normally
AT-46	Leave a session quarantined and read the heartbeat	REC-08, REC-09	Stuck bytes reported centrally and visible in the operator view
AT-47	Send a trigger with demographics and a previous prescription	CRI-01, CRI-02	Both stored; recording unaffected
AT-48	Withhold the clinical record entirely for one encounter	CRI-04, CRI-10	Recording completes; reconciliation reports the gap next night
AT-49	Deliver a record failing schema validation	CRI-07	Stored unmodified in quarantine and reported; nothing partially ingested
AT-50	Deliver the same record twice	CRI-09	One row, not two
AT-51	Grep the archive paths, filenames and chain for the patient's name	CRI-08, DAT-02	Not present in any of them
AT-52	Record a third encounter for one patient	DBA-07, DBA-08	All three retained; the views return the correct current and previous
AT-53	Two encounters for one patient on the same day	DBA-09	<code>previous_prescription</code> returns the earlier one, never the encounter in progress
AT-54	Insert a female-only field for a male patient, and omit a required one	DBA-06	Rejected by a database constraint, not by application code
AT-55	Query recordings for one clinic, clinician and date	DBA-02, DSH-02	Served by index; recording located without a filesystem search
AT-56	Open the dashboard as an operations role	DSH-06	Volumes, integrity and reconciliation visible; no patient name or clinical content reachable
AT-30	Deploy the backend during an active recording	NFR-03	No interruption; no lost segment
AT-31	Send a 128 KB frame	IF1 transport	Refused, connection preserved
AT-32	Send a message with an unknown extra field	NFM-05	Ignored; command succeeds
AT-33	Full pipeline: consultation → archive	ARC-01..05, SES-05	Correct filename; chain verified; purge receipt issued; local audio deleted after grace

12.2 Exit criteria for the pilot

The integration is accepted when:

1. AT-01 through AT-33 pass on a clinic-representative machine.
2. Twenty consecutive real consultations at one clinic archive with verified

chains and correct filenames.

1. Zero incidents in which a doctor was prevented from working by AIMscribe.
2. One induced-power-loss trial and one internet-outage trial pass in the field,

not only on the bench.

1. Every **OD-*nn*** is closed or explicitly deferred in writing.

12.3 Traceability

Requirement group	Verified by
INV	AT-02, AT-04, code inspection
ENR	AT-21 – AT-26
GRT	AT-04, AT-07, AT-08, AT-28
IF1	AT-01, AT-05, AT-06, AT-27, AT-31, AT-32
CAP	AT-34, AT-35, AT-37, commissioning measurement
SEG	AT-38, AT-39, AT-01
LVL	AT-36, AT-37
SPL	AT-16, AT-17, AT-20
CHN	AT-19, AT-33
UPL	AT-16, AT-18
SES	AT-10, AT-33
GAT	AT-09 – AT-12
UIX	AT-13 – AT-15
BKD , ARC	AT-03, AT-30, AT-33
NF*	AT-29, AT-30, plus measurement against §9

13. Delivery plan and division of work

13.1 What already runs, and what does not

This document is a design baseline, not a description of shipped software. The split matters for planning and is stated here rather than discovered in a meeting.

Working today	Still to be built	Owner
WebSocket transport, origin + Host + loopback admission	—	AIMS (done)
<code>start</code> , <code>stop</code> , <code>pause</code> , <code>resume</code> , <code>status</code> , <code>doctors</code>	—	AIMS (done)
Grant verification: Ed25519, 60 s, single-use <code>jti</code>	—	AIMS (done)
Enrolment, device tokens, revocation, audit log	—	AIMS (done)
Hash chain, encrypted spool, presigned upload, read-back verification	—	AIMS (done)
Seamless patient-to-patient handover	—	AIMS (done)
—	The trigger payload of §6.1.3 (today the page mints its own grant)	Joint
—	<code>POST /grant/mint</code> on the backend — does not exist	AIMS
—	<code>consultation_complete</code> and the gate (§7.7)	AIMS
—	The <code>request_id</code> / <code>status</code> / <code>code</code> envelope	AIMS
—	<code>202 RECORDING_PROVISIONAL</code> ; clinic-mismatch refusal	AIMS
—	The recording-control overlay (§7.8)	AIMS
—	The durability checkpoint (<code>SRS-SPL-10. .12</code>)	AIMS
—	Database production hardening	AIMS

Every item is AIMS LAB's except one. **The trigger payload is the single place both sides change at once**, which means §6.1.3 and §5.3 must ship together: an integration written against the new five-field trigger will otherwise be talking to an agent that still expects a signed grant.

13.2 CMED's work, itemised

#	Task	Estimate
1	Open and maintain the WebSocket with backoff	0.5 d
2	Send the trigger on the patient-details action	0.5 d
3	Send <code>consultation_complete</code> on prescription build	0.25 d
4	Handle replies by <code>code</code> ; log and ignore failures	0.5 d
5	Optional recording indicator driven by events	0.5 d
6	Add <code>connect-src</code> to CSP	0.1 d
7	Supply exact production and staging origins	0.1 d
8	Joint integration testing	1 d
	Total	2–4 developer-days

Appendix B is a reference client that covers items 1, 2, 3 and 4.

13.3 What AIMS LAB brings

Working agents on enrolled machines, a running backend, the archive, the AI pipeline downstream, installers and support, all keys and their rotation, server capacity and its cost, and this specification.

13.4 Suggested sequence

Phase	Content	Gate to the next phase
0	Agree origins, clinic mapping, identifier stability, and the OD list	Written agreement
1	AIMS: build <code>/grant/mint</code> , the trigger payload, the reply envelope	<code>AT-01</code> – <code>AT-08</code> pass on a bench
2	CMED: implement items 1–4 against a staging agent	<code>AT-27</code> , <code>AT-32</code> pass
3	AIMS: gate + overlay	<code>AT-09</code> – <code>AT-15</code> pass
4	Joint: one clinic, one week	§12.2 exit criteria
5	Fleet rollout, clinic by clinic	Per-clinic sign-off

14. Open decisions

ID	Decision	Needed from	Blocks	Consequence of delay
OD-01	Exact production and staging origins for the allowlist	CMED	Phase 2	Nothing can connect
OD-02	Is <code>patient_id</code> guaranteed stable for a patient over time?	CMED	Phase 0	Recordings for one patient split into unrelated sets
OD-03	Clinic mismatch: refuse (recommended) or warn-and-record for the pilot?	Joint	Phase 1	Default is refuse, per <code>SRS-GRT-10</code>
OD-04	What share of consultations end without a prescription being built?	CMED / clinical	Phase 3	Decides whether the gate also needs an inactivity timeout
OD-05	Mean consultation duration — measured, not estimated	Clinical	Phase 4	\$10 sizing scales linearly with it
OD-06	Raw-audio retention period	AIMS LAB	Hardware purchase	11 TB/year; the single largest cost driver
OD-07	Stop/pause reason vocabulary, and which reasons count as abnormal	Clinical	Phase 3	The overlay cannot be built without the list
OD-08	Does the overlay need a supervisor-name field for long pauses?	Clinical	Phase 3	Affects <code>SRS-UIX-10</code>
OD-09	Durability checkpoint interval	Measurement	Phase 3	Must come from the fsync benchmark, not from a guess
OD-10	Backend hosting: cloud or AIMS LAB server	AIMS LAB	Phase 4	\$10.3 vs \$10.4
OD-11	Whether the existing speakerphone can meet §7.1a, or the microphone must be replaced	Measurement	Commissioning	Decided by <code>AT-34</code> / <code>AT-35</code> , not by opinion
OD-12	Capture at the device's native 48 kHz rather than 44.1 kHz	AIMS LAB	Phase 1	Removes an OS resample and makes the 16 kHz recognition decimation exact; costs 8.8% storage

15. Appendices

Appendix A — Response codes

A.1 start

status	code	Meaning	CMED action
200	RECORDING_STARTED	Recording; <code>session_id</code> returned	Show the indicator
202	RECORDING_PROVISIONAL	Capturing; authorisation still in flight	Show the indicator; a later event confirms
400	MISSING_FIELD	A mandatory field is absent	Fix the payload — a build-time bug
400	INVALID_IDENTIFIER	An identifier fails the pattern	Fix the identifier
400	CONSENT_REQUIRED	<code>consent_obtained</code> not true	Do not retry until consent is recorded
401	AUTHORISATION_FAILED	Grant minting or verification failed	Log; do not retry automatically
401	CLINIC_MISMATCH	Clinic does not match the device's enrolment	Log and raise with AIMS LAB; do not retry
403	ORIGIN_NOT_ALLOWED	Origin not allowlisted	Configuration issue; contact AIMS LAB
404	DOCTOR_NOT_AT_CLINIC	Doctor unknown at this clinic	Log; check the register
409	GATE_NOT_ARMED	A consultation is open and unfinished	Expected. Do nothing; keep working.
409	SESSION_ALREADY_ACTIVE	This exact session is already recording	Ignore — usually a double click
423	DEVICE_NOT_ENROLLED	This PC is not enrolled	Contact AIMS LAB
503	AGENT_NOT_READY	Agent starting or recovering	Retry once after a short delay
—	(no connection)	Agent absent or stopped	Log and carry on. Never block the doctor.

A.2 consultation_complete

status	code	Meaning	CMED action
200	GATE_ARMED	Accepted; the next trigger may hand over	None
200	GATE_ALREADY_ARMED	Already armed; idempotent	None
400	MISSING_FIELD	<code>patient_id</code> absent	Fix the payload
409	NO_ACTIVE_SESSION	Nothing is recording	Ignore
409	PATIENT_MISMATCH	Names a patient other than the open session's	Log — worth investigating

A.3 Error frame

```
{
  "event": "error",
  "command": "start",
  "request_id": "cmed-7f3a2b91",
  "status": 409,
  "code": "GATE_NOT_ARMED",
  "message": "The current consultation has not been completed yet.",
  "timestamp": "2026-08-25T10:14:32+06:00"
}
```

Branch on `code` . Never on `message` . Message text is shown to doctors and will be reworded without notice; `code` is contractual.

Appendix B — Reference client

```
/**
 * Minimal AIMScribe client for the CMED page.
 *
 * Every public method resolves rather than throws. A recorder that fails must
 * never stop a doctor seeing a patient – SRS-IF1-17.
 */
type Ack = { status: number; code: string; data?: Record<string, unknown> };

export class AIMScribeClient {
  private ws: WebSocket | null = null;
  private backoff = 1000;
  private readonly pending = new Map<string, (a: Ack) => void>();

  constructor(private readonly onEvent: (e: any) => void = () => {}) {}

  async connect(): Promise<void> {
    // Probe first: this distinguishes "not installed" from "failing".
    try {
      const probe = await fetch('http://127.0.0.1:5050/health', {
        signal: AbortSignal.timeout(1500),
      });
      if (!probe.ok) return this.retry();
    } catch {
      return this.retry(); // agent absent – entirely normal
    }

    const ws = new WebSocket('ws://127.0.0.1:5050/ws');
    ws.onopen = () => { this.ws = ws; this.backoff = 1000; };
    ws.onclose = () => { this.ws = null; this.retry(); };
    ws.onerror = () => { /* onclose follows; nothing to do */ };
    ws.onmessage = (m) => {
      let msg: any;
      try { msg = JSON.parse(m.data); } catch { return; }
      const resolve = msg.request_id && this.pending.get(msg.request_id);
      if (resolve) { this.pending.delete(msg.request_id); resolve(msg); }
      else { this.onEvent(msg); }
    };
  }

  private retry(): void {
    setTimeout(() => this.connect(), this.backoff);
    this.backoff = Math.min(this.backoff * 2, 30000);
  }

  private send(payload: object): Promise<Ack> {
```

```

    if (!this.ws || this.ws.readyState !== WebSocket.OPEN) {
      return Promise.resolve({ status: 0, code: 'AGENT_UNAVAILABLE' });
    }
    const request_id = `cmed-${crypto.randomUUID().slice(0, 8)}`;
    return new Promise((resolve) => {
      this.pending.set(request_id, resolve);
      // A reply that never arrives must not leak a pending entry.
      setTimeout(() => {
        if (this.pending.delete(request_id)) {
          resolve({ status: 0, code: 'AGENT_TIMEOUT' });
        }
      }, 3000);
      this.ws!.send(JSON.stringify({ ...payload, request_id }));
    });
  }

  startConsultation(t: {
    patient_id: string; doctor_id: string; clinic_id: string;
    start_time: string; date: string;
    consent_obtained: boolean; consent_method?: string; patient_name?: string;
  }): Promise<Ack> {
    return this.send({ command: 'start', trigger: t });
  }

  consultationComplete(patient_id: string): Promise<Ack> {
    return this.send({
      command: 'consultation_complete',
      patient_id,
      occurred_at: new Date().toISOString(),
    });
  }
}

```

The two call sites.

```
// When the doctor opens a patient's details or history.
aimscribe.startConsultation({
  patient_id: patient.id,
  doctor_id: session.doctor.id,
  clinic_id: session.clinic.id,
  start_time: new Date().toISOString(),
  date: todayLocalISODate(),
  consent_obtained: patient.consentOnFile,
  consent_method: 'verbal_at_reception',
}).catch(() => {}); // SRS-IF1-17, in three characters

// When Build Prescription succeeds.
aimscribe.consultationComplete(patient.id).catch(() => {});
```

Those two `.catch(() => {})` calls are the most important lines in the integration. They are `SRS-IF1-17` expressed in code: whatever goes wrong in AIM Scribe, the doctor keeps working.

Appendix C — API reference

Two surfaces. CMED implements only the first, and only two of its commands.

C.1 Partner → agent, on the consulting-room PC

`ws://127.0.0.1:5050/ws` · JSON frames · 64 KB maximum · `Origin` mandatory

Command	Sent when	Required fields	Success
<code>start</code>	Doctor opens patient details	<code>patient_id</code> , <code>doctor_id</code> , <code>clinic_id</code> , <code>start_time</code> , <code>date</code> , <code>consent_obtained</code>	<code>200 RECORDING_STARTED</code>
<code>consultation_complete</code>	Prescription built	<code>patient_id</code> , <code>occurred_at</code>	<code>200 GATE_ARMED</code>
<code>status</code>	Any time	—	current state
<code>pause</code> · <code>resume</code> · <code>stop</code>	Not used by CMED	—	—

`GET http://127.0.0.1:5050/health` is unauthenticated, has no side effects, and is the correct way to ask whether the agent is present. Full outcome tables are in Appendix A.

C.2 Agent and worker → AIMS LAB backend

Internal to AIMS LAB. Listed so a partner's reviewer can see the whole chain of custody; CMED calls none of it.

Method	Path	Auth	Purpose
POST	/api/v2/device/enroll	enrolment token	Exchange a one-time token for a device identity
POST	/api/v2/grant/mint	device token	Authorise one consultation (<i>to be built, §5</i>)
POST	/api/v2/session/open	device token	Open a session, submit chain entry 0
POST	/api/v2/segment/authorize	device token	Presigned PUT for one segment, 300 s
POST	/api/v2/segment/commit	device token	Declare uploaded, with SHA-256; server re-reads and re-hashes
POST	/api/v2/session/pause · /resume	device token	Supervised pause with its reason
POST	/api/v2/session/close	device token	Close and submit the chain tail
GET	/api/v2/session/{id}/receipts	device token	Collect purge receipts
POST	/api/v2/heartbeat	device token	Liveness, spool depth, version, level metrics
GET	/api/v2/doctors	device token	Typing suggestions, own clinic only
GET	/api/v2/archive/pending	worker key	Sessions ready to archive
POST	/api/v2/archive/complete	worker key	Report an archived session
POST	/api/v2/admin/hospital · /doctor	admin key	Register a clinic or a doctor
POST	/api/v2/admin/enrollment-token	admin key	Mint a single-use enrolment token
POST	/api/v2/admin/device/{id}/revoke	admin key	Cut off a lost machine
GET	/api/v2/admin/alerts	admin key	Unresolved integrity alerts

Three separate bearer credentials — `X-Device-Token` , `X-Worker-Key` , `X-Admin-Key` — which are not interchangeable (`SRS-IF2-02`).

C.3 Testability

SRS-API-01 [M, D, AIMS] A Postman collection shall be published covering every route in C.2, with an environment file holding base URL and credentials as variables, so no secret is committed with the collection.

SRS-API-02 [M, D, AIMS] The collection shall include a working end-to-end sequence — enrol, mint, open, authorize, commit, close, receipts — that a reviewer can run against a staging deployment without reading code.

SRS-API-03 [M, I, AIMS] The backend shall serve OpenAPI at `/openapi.json` and interactive documentation at `/docs` , so the collection can be regenerated rather than hand-maintained.

SRS-API-04 [S, D, AIMS] A partner-facing WebSocket test page shall be provided — a single HTML file that connects to `127.0.0.1:5050` , sends a trigger and a flag, and prints the replies — because Postman does not exercise the loopback WebSocket surface CMED actually implements.

SRS-API-04 matters more than the Postman collection for CMED specifically: their entire integration is the WebSocket, which Postman is the wrong tool for.

Appendix D — Requirement index

Prefix	Area	\$
INV	Identity invariants	2.2
USR	User constraints	2.3
TOP	Topology and trust boundaries	3.4
ENR	Enrolment	4
GRT	Grants	5
IF1 / IF2 / IF3	Interfaces	6
CAP	Capture and the capture path	7.1, 7.1a
LVL	Speech-level monitoring	7.1b
SEG / SPL / CHN / UPL	Segmentation through upload	7.2–7.5
REC	Recovering unverified audio	7.5a
SES / GAT / UIX	Sessions, gate, overlay	7.6–7.8
BKD / ARC	Backend and archive worker	7.9–7.10
DAT	Data	8.1–8.5
CRI	Clinical record ingestion	8.6
DBA	Database architecture	8.7
DSH	Operational dashboard	8.8
NFP / NFC / NFR / NFS / NFO / NFM / NFU	Non-functional	9

Appendix E — Document history

Version	Date	Change
1.0	25 August 2026	First baseline for the CMED integration meeting
1.7	9 September 2026	Quarantined material is delivered like any other, on a drainage scheduler; nothing remains on a workstation and no path requires a person to touch a clinical PC (SRS-REC-10 – 14 , SRS-SPL-09/13/14). One database with two schemas replaces the two-database split. The search problem re-diagnosed: the indexes already existed, so SRS-DBA-22 – 25 address the split catalogue, autosuspend and substring search instead.
1.6	9 September 2026	Clinical record ingestion (§8.6), database architecture (§8.7) and the operational dashboard (§8.8) added; §8.1 reversed and the reversal recorded, with SRS-DAT-15/16 on consent and access. SRS-SPL-13/14 close the one case where quarantined material could consume the spool unreported.
1.5	5 September 2026	Prepared for submission. The six diagrams that were mermaid source are now drawn figures, so the document is complete in print; figures renumbered into order of appearance.
1.4	25 August 2026	Corrected §7.5a: the agent already re-verifies every segment locally before upload, so local damage is caught before it leaves the PC and a server-side mismatch is almost never damaged audio. Added §7.5a.1 enumerating all five quarantine triggers and which of them involve the recording at all — only one does, and it never reaches the server.
1.3	25 August 2026	Added §7.5a SRS-REC-01 – 09 and tests AT-42 – AT-46 : local re-verification before giving up, bounded retry to a fresh object key, per-segment rather than per-session quarantine, an un-quarantine path, preservation of unverifiable audio outside the chain, and central visibility of stuck local audio.
1.2	25 August 2026	Figures regenerated from a layout pass that fails the build on any label collision. Added Appendix C (API reference and testability, SRS-API-01 – 04) and SRS-IF1-16/18 : CMED renders nothing; all doctor messaging is delivered by our overlay.
1.1	25 August 2026	Sizing corrected to 14 rooms. Added the capture path (§7.1a), speech-level monitoring (§7.1b), the original/derivative rule (§8.5), the clinic register (§4.6), SRS-ENR-21 , SRS-SEG-06/07 , tests AT-34 – AT-41 , and figures 1–3.

Questions, corrections and disagreements are welcome and expected. A specification that survives its first contact with an implementer unchanged was probably not read carefully enough.

AIMS LAB · Independent University, Bangladesh